3 September 2023

Hitachi Energy Pty Ltd.

Hitachi Energy Pty Ltd Technical Offer in support of the proposal for a Plant Settlement System (PSS).

Attention:

Larsen & Toubro on behalf of NAWWAR Renewable Energy Company (Arrass PV Project Site)

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1. Executive summary

Background and Objective

To support the performance testing and financial settlement for the Arrass Independent Power Plant utilizing Photovoltaic Energy Generation. The Saudi Power Procurement Company (Offtaker) has mandated need for a Plant Settlement System (PSS) delivering Performance Monitoring and Back Office Settlement calculation and invoicing capabilities. Hitachi Energy proposes to implement its SettlementTracker and Forecast module, delivered as an on-premises installation, to support the EPC's PSS requirements.

The objective of the PSS is to facilitate the managing the Performance & Testing of the Arrass Solar PV plant and the settlement of net electricity generation and consumption charges between the IPP and the Saudi Arabian Power Procurement Company (SPPC).

2. Introduction to Hitachi Energy

2.1. Our purpose

‘Hitachi Energy – Advancing a sustainable energy future for all’

We are advancing the world’s energy system to be more sustainable, flexible, and secure. As the Energy technology leader, we collaborate with customers and partners to enable a sustainable energy future – for today's generations and those to come.

2.2. About Hitachi Energy

Hitachi Energy is a global technology leader with a combined heritage of almost 250 years, employing around 38,000 people in 90 countries. Headquartered in Switzerland, the business serves customers in the renewables, utility, industry and infrastructure sectors with innovative solutions and services across the value chain. Together with customers and partners, Hitachi Energy Hitachi Energy’s technologies and enables the digital transformation required to accelerate the energy transition towards a carbon-neutral future.

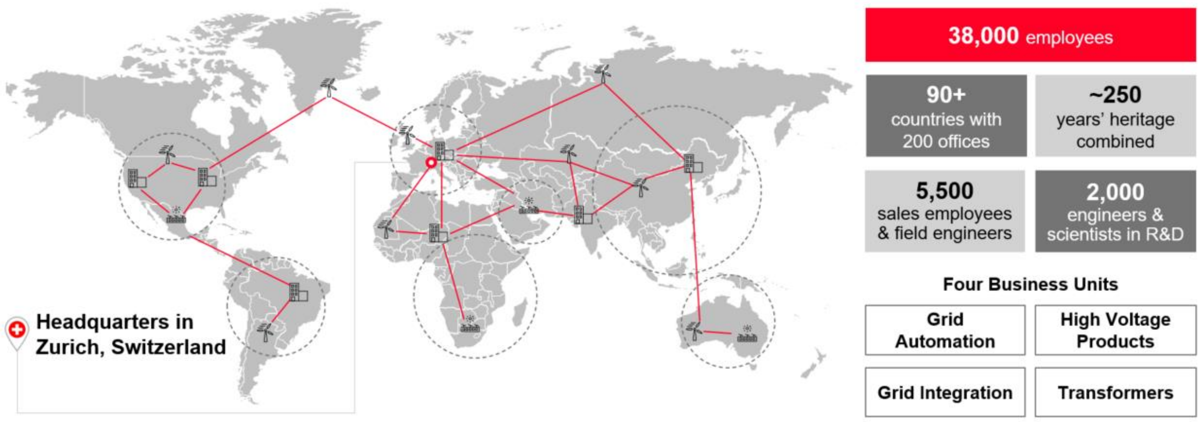

Figure 1: Overview of Hitachi Energy

Hitachi Energy serves the following markets:

- Renewables – Accelerating renewable integration with a strong installed base.
- Utilities – Partnering with utilities for 130 years from generation to transmission and distribution.
- Industries – Supporting industrial customers to electrify the entire energy value chain.
- Transportation – Enabling society to meet sustainable mobility demands in air, land, water and rail.
- Data centres – Providing the data-centre industry with reliable power connection and eco-efficient solutions.
- Smart Life – Advancing sustainable energy for industry and society with solutions that reduce waste and CO_2 footprint.

Grid Automation

With over 30 years of experience and 6,000 employees, Grid Automation is one of the main four Hitachi Energy divisions:

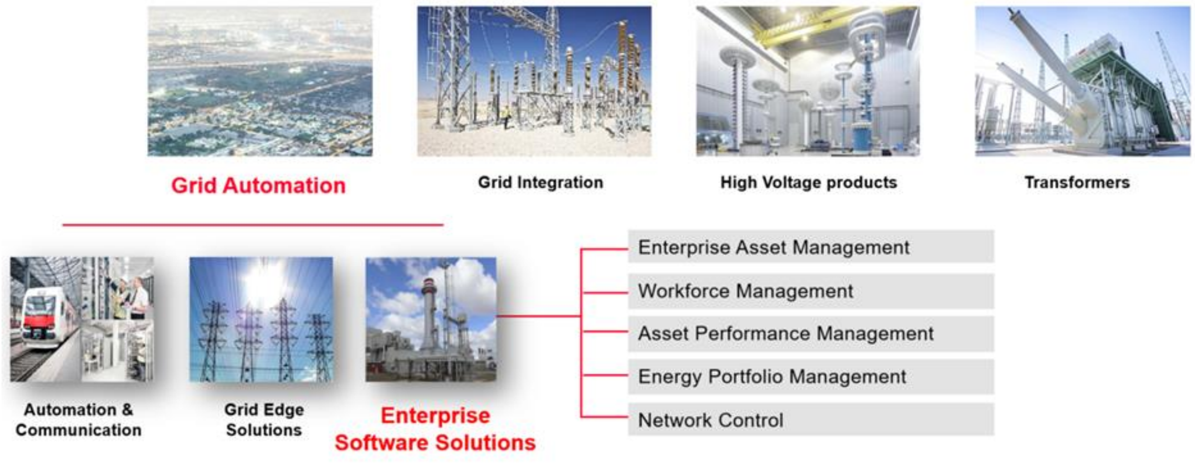

Figure 2: Overview of Grid Automation division

Enterprise Software Solutions is part of the Grid Automation division.

Enterprise Software Solutions (ESS)

From renewable developers to urban rail operators, from energy generation, gas and water, transmission, and distribution to mining and industrial companies, for over 1,100 customers around the world, the Enterprise Software Solutions (ESS) arm of Hitachi Energy is dedicated to delivering a better model of enterprise business solutions – one that takes an integrated whole-systems approach to balance the enterprise asset risk, performance, and cost.

Our software is underpinned and influenced by Hitachi Energy's collective history across multiple asset-intensive industries. The breadth of our mature software solutions includes Enterprise Asset Management (EAM), Asset Performance Management (APM), Energy Portfolio Management (EPM), Field Mobility, SCADA EMS, Smart Transformers, and Digital Substations.

As the only industrial enterprise software solution provider with a combined integration of information technology (IT) systems with operational technology (OT) systems (IT/OT) offering, Hitachi Energy and our selected partner environment work seamlessly to execute best-of-breed solutions for your most critical business objectives. The ESS product group has been providing software, intelligence and services that serve the unique needs of essential industries.

Energy Portfolio Management (EPM)

Hitachi Energy has been partnering with essential industries for more than 120 years, providing software that serves the unique needs of the energy industry for over 40 years through its Energy Portfolio Management (EPM), of which the TRMTracker solution is a key part.

EPM provides the energy industry with integrated software solutions, energy markets intelligence and strategic advisory services that enable industry professionals to make better economic and strategic decisions. This enables system planners, regulators, energy market participants (including developers and investors) and market operators in assessing infrastructure investments as well as providing critical information and systems to support economic operations of energy assets.

Our offering is based on global market experience across the Middle East, Europe, the Americas, and the Asia Pacific regions. EPM supports over 500 customers around the world in improving their strategic planning, investments, trading, and operations decisions.

EPM’s portfolio comprises the following integrated offerings for its customers:

- Energy Trading and Risk Management solutions: To automate tasks and business processes supporting the entire trade cycle from deal capture and contract management to pricing and complex fees, trade confirmations, portfolio management and valuations, risk controls, collateral, and credit management, to settlement, and regulatory compliance reporting.
- Energy Market Analysis Software: Capacity expansion, market simulation, and portfolio optimisation.
- Market Intelligence Data: Electricity system data (power plants, transmission lines and constraints), historical, real-time, and forecasted critical energy information/data.
- Energy Market Outlook Reports Country/market-specific analysis projecting system and market development over the next 25 years horizon for 28 European countries, North America, and Asia-Pacific.
- Strategic Advisory Services: Electricity system/Market studies, integration of renewables, due diligence/feasibility of energy assets and risk assessment.

EPM’s mission is to support critical energy system planning, asset investment/divestment decisions and to support efficient energy operations across the planning, forecasting, trading, plant operation and portfolio optimisation.

Figure 3: Overview of EPM portfolio awards

2.3. Why Hitachi Energy

Hitachi Energy has prepared this technical and functional ability overview that provides a strong value in alignment with the requirements outlined by ARRASS.

Hitachi Energy’s offering includes numerous combinations of strength and delivery, as listed below.

- Proven solution: TRMTracker has been selected by forward-thinking companies and voted by Energy Risk Software Survey as Best Front Office software, Best Middle Office Software, and Best Integration Capability in the 2021 rankings.
- Proven track record: Hitachi Energy have implemented similar solutions for other clients with similar requirements. Hitachi Energy is confident in meeting Arrass’ project objectives.

Hitachi Energy

- Business knowledge: Hitachi Energy professionals understand the utility industry and can work closely with ARRASS to design the solution, implement the solution, and enhance the solution in the future to drive the greatest return for ARRASS investment.
- Market knowledge: From commercial operations, trading, retail operations, unit commitment, ISO communications, long and short-term forecasting, asset management, mobile workforce, to customer service Hitachi Energy has the knowledge, experience, and resources.
- Partnering: Hitachi Energy views ARRASS as a partner. We understand your business as described in the Tender and believe that our interests in this area of business are aligned. Hitachi Energy is committed to working with ARRASS to deliver a solution that drives the greatest value for ARRASS.
- Worldwide software and consulting services: Hitachi Energy provides software and consulting services worldwide involving all aspects of utility planning and operations. For more than 40 years, Hitachi Energy has provided professional consulting services and software to serve the critical utility functions of integrated resource planning, market price forecasting, resource evaluation and planning, trading and settlement, and electric transmission economic analysis, customer demand forecasting, system and grid operational management, and individual plant operations and performance optimisation.
- Software training: Hitachi Energy has a well-established training programme for the users of its applications/software. This training is conducted by highly experienced consultants who have in-depth knowledge of the application's implementation, operation, and energy markets.
- Market reports from a trusted advisor with a unique offering: EPM is a trusted advisor on power markets and power systems, and energy portfolio risk. Our advisory services help our clients turn their energy markets strategies into actionable results. Our unique strength in providing a combination of both software and advisory services means we can provide market-leading solutions.
- Enabling a stronger, smarter, and greener grid: Hitachi Energy is committed to providing the technology and services to support the integration of renewables. It integrates grids with new forms of energy, providing a full service from consulting, generation and connection to transmission, monitoring, and control, as well as maintenance and optimisation. Hitachi Energy has the tools, the experience, the resources, and the track record to advise and assist the client in the efficient development of a long-term generation development strategy in modern electrical systems.

3. Objective

To support the performance testing and financial settlement for the Arrass Independent Power Plant utilizing Photovoltaic Energy Generation. The Saudi Power Procurement Company (Offtaker) has mandated need for a Plant Settlement System (PSS) delivering Performance Monitoring and Back Office Settlement calculation and invoicing capabilities. Hitachi Energy proposes to implement its SettlementTracker and Forecast module, delivered as an on-premises installation, to support the EPC's PSS requirements.

The objective of the PSS is to facilitate the managing of the Performance & Testing of the Arrass Solar PV plant and the settlement of net electricity generation and consumption charges between the IPP and the Saudi Arabian Power Procurement Company (SPPC).

3.1. Scope

3.1.1. Project Activities

The following activities are included within the scope of this project:

- Project Management (Project management includes managing Hitachi Energy resources and tasks)
- Detailed project plan for execution phase
- Design documentation as required.
- Implement/Configure required interfaces.
- Necessary knowledge transfer to administrative functional users
- Factory Acceptance Testing pre-approved by EPC Stakeholders
- Functional training
- Assistance in creating UAT test scripts and test plan. Note that EPC is responsible for providing test scenarios.
- Assistance in functional UAT testing. Note that the EPC is responsible for running UAT testing.
- Setup required business processes in SettlementTracker and Forecast module (refer to section 4.5.3.) for further details).

3.1.2. Functional Requirements

Hitachi Energy shall provide:

- a PSS installed on premise at both the Arrass generating plant site (Online) and the Saudi Power Procurement Company's (SPPC's) data center (Offline).

- configure a "Replica Database" installed on premise within the DMZ at the SPPC's data center.

- Delivery of data from the Arrass Meter Data Acquisition System (MDA), PV SCADA system and Plant Settlement System (PSS). The Replica DB shall each provide a database schema configured as described in this document in compliance with SPPC Specification for receiving two distinct data types from the PSS, MDA and PV SCADA systems.

The scope of the PSS provided shall include as follows:

- Calculations included within Schedule 9 of the PPA - Calculation of Payment.
- Business Processes/Workflows included within Schedule 13 of the PPA – Invoicing and Payment Procedures.
- Calculations included within Schedule 17 of the PPA – Deemed Energy Calculation – Methodology.
- MDA will be responsible for pushing data received from the PV SCADA and Online PSS on an hourly or 6-minute frequency basis as appropriate for hourly and six-minute meter and PV SCADA data updates by means of ODBC over an encrypted transportation layer.
- Pushing the 'online results' and 'offline results' data to the CSS Database via ODBC.

This document is organized into the following additional sections:

1. **Functional and Non-functional Requirements:** This section contains functional and non-functional requirements and scope. This section also contains static data definition, interfaces, data migration requirements and strategies, technical requirements, and gaps.

2. **Enhancements Assessment and Gap Closure Plan:** This section documents any enhancements/gaps identified and resolution plan.

4. Functional and Non-Functional Requirements

This section defines functional, non-functional, and other requirements for the implementation.

The Arrass Solar PV Power Purchase Agreement governs the requirements for performance monitoring and settlement charges between the PPA counterparties IPP (and the Offtaker (Saudi Power Procurement Company – SPPC).

The Arrass PPA Settlement System comprises the following four installations,

- Plant Settlement Systems
  (1) the Online Plant Settlement System (Online) hosted at the Independent Power Producer (IPP) Arrass Solar PV plant
  (2) the Offline Plant Settlement System (Offline) hosted at the Saudi Power Procurement Company's (SPPC) datacenter
  (3) the Central Settlement System provided by the SPPC. The Online and Offline system each comprise a further two logical sets of functional requirements.
  (4) Meter Data Acquisition solution

The Plant Settlement Systems include the functional capabilities to perform the roles of:

- Solar Performance Monitoring (SPM) (a.k.a. PPM - Power Performance Model); and the
- Plant Accounting and Settlement System (PSS). (a.k.a. TCM - Tariff Calculation Model)

The Online system shall manage the settlement of a single PPA in respect of the Arrass Solar PV plant, while the Offline system shall receive the same input data as the Online system and may perform both shadow settlement calculations and scenario calculations.

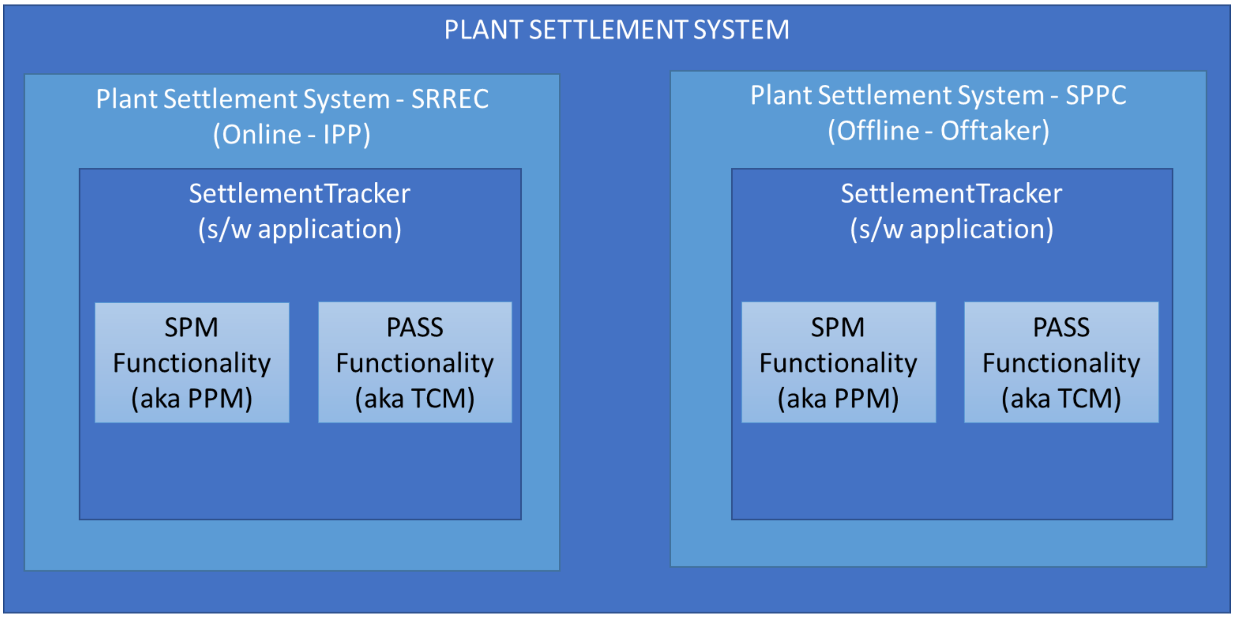

Figure 1: Plant Settlement System Component Naming Conventions

Hitachi Energy's SettlementTracker and Forecast Module shall be deployed on premises hosted on the servers of the IPP and the SPPC (the Offtaker) and configured to deliver these functional requirements to each counterparty.

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The following Logical Data Flow Diagram (Figure 2) illustrates the key entities, software applications and logical data flow relationships within the scope of this project and identifies which systems and associated interfaces that are not within the scope of this project.

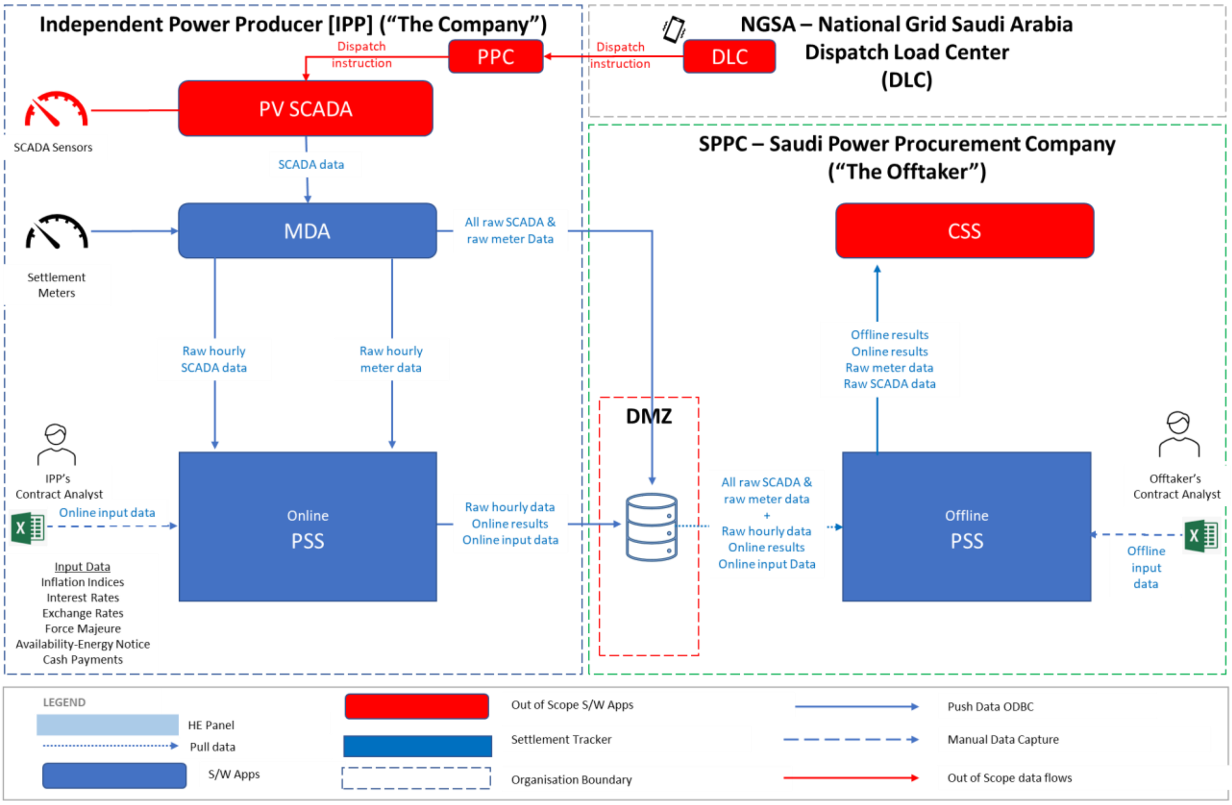


Figure 2: Arrass Plant Settlement System Logical Data Flow Diagram

The SPPC Interface Specification V7 refers to 'Raw data/input' in sections 6 Offline and Online PSS and 7.2 Data Input Guidelines. Data/Interfaces (Others)

Integration Requirements

4.1. Interfaces

There shall be three data interfaces with the PSS:

1. The Meter Data Acquisition (MDA) system.
2. The Photovoltaic Supervisory Control and Data Acquisition (PV SCADA) system.
3. The SPPC Central Settlement System (CSS)

The MDA system shall interface with the PV SCADA system using the 104 communications protocols and the MDA shall pass through the subset of required PV SCADA data to the Online PSS via an ODBC interface.

The MDA system shall provide the ODBC interface mechanism for both the online and offline PSS instances and shall manage the transfer of raw meter data and PV SCADA data, hourly meter and PV SCADA data, online and offline settlement results data, online and offline input data and forecast generation data between the online PSS, Replica DB, offline PSS and CSS.

The following diagram Figure 4 shows the internal IPPIPPSPPC system's interfaces (i.e., interfaces between those software systems operated by either the IPP or SPPC)

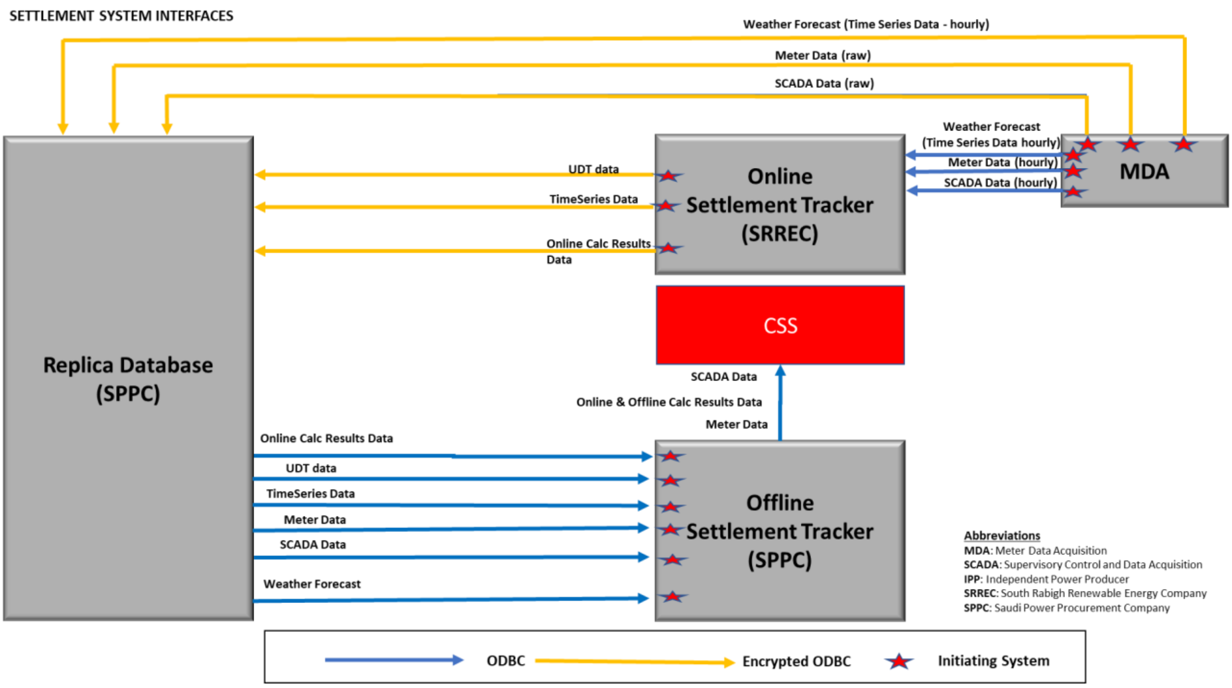


Figure 3: Settlement Tracker Data Interfaces

With respect to the above diagram Figure 3 Settlement Tracker Data Interfaces:

User Defined Table (UDT) data shall include the following.

- Monthly Coefficient A, B and C data
- Weather data actual
- Sunrise and Sunset info
- Guaranteed Performance Ratio
- Projected Net Electrical Energy
- Availability Energy Notice

Time Series data shall include the following.

- Weather forecast data (SCADA)
- Outcome meteorological data (SCADA)
- Inflation Indices
    - US Producer Price Index (TBC following availability of PPA Contract)
    - KSA Consumer Price Index (TBC following availability of PPA Contract)

- Interest Rates
    - SAIBOR (TBC following availability of PPA Contract)
    - SOFR (TBC following availability of PPA Contract)
- Exchange Rate
    - USD – SAR (TBC following availability of PPA Contract)
- Cash Received
- Generation Forecast

4.1.1. MDA & PV SCADA

The communication interfacing for transferring data from the MDA to the (Online system or IPP's) Settlement System and SPPC's CSS shall be via ODBC.

The same ODBC communication protocol will be used for (Offline or Offtaker's) Central Settlement System at SPPC via MDA.

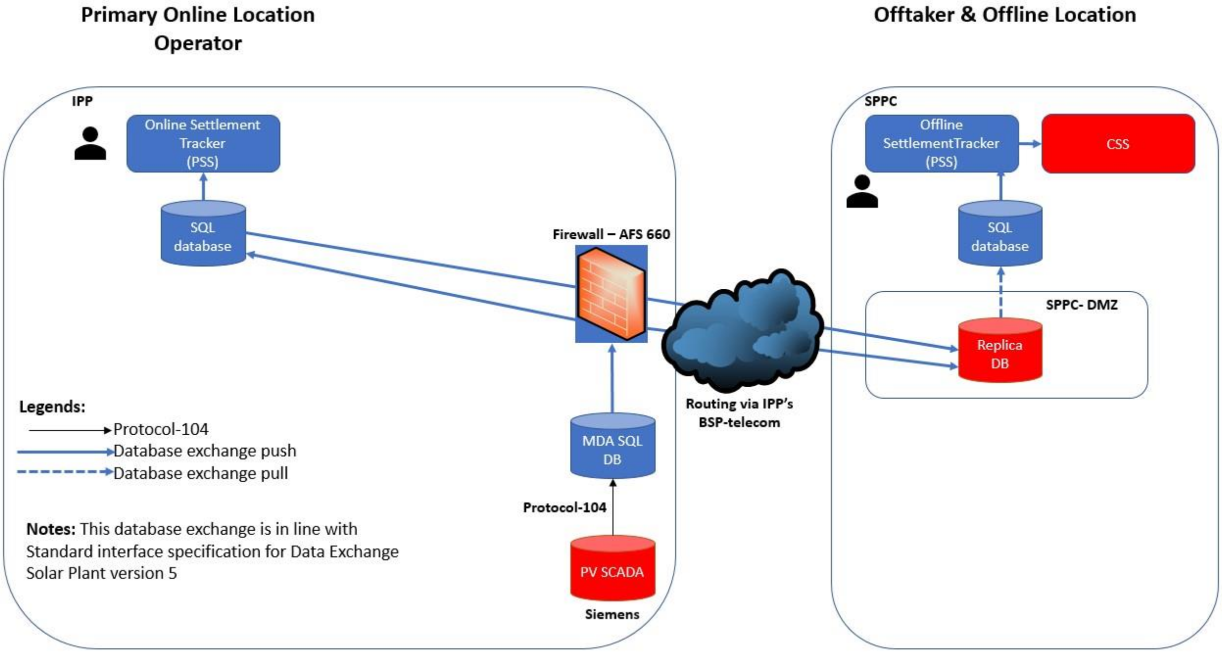


Figure 4: Interaction of PSS/SPM with MDA & PV SCADA and with the settlement infrastructure for synchronization at CSS at SPPC premises

4.1.2. Open Database Connectivity

Microsoft Open Database Connectivity (ODBC) is an application programming interface (API) to access data from a different database management system (DBMS). ODBC is designed for relational data stores.

ODBC enables the SQL Server Database Engine to read data from the remote data sources and execute commands against the remote database servers (for example, OLE DB data sources) outside of the instance of SQL Server. Typically, ODBC is configured to enable the Database Engine to execute a Transact-SQL statement that includes tables in another instance of SQL Server, or another database product such as Oracle.

4.1.3. Plant Settlement System (PSS)

Online PSS

The PSS online system shall receive from the MDA, hourly meter and PV SCADA data including the weather forecast. This data is an hourly subset of the raw 5-minute and less granularity data exported by the MDA.

The online calculated Settlement Results shall be exported from the PSS monthly with hourly data granularity.

The online input data shall be exported from the PSS daily with hourly data granularity.

Offline PSS

The PSS offline system shall each day at a scheduled time pull via ODBC up to the previous 31 days updated online input data from the Replica DB.

The PSS offline system shall each month also push calculated offline settlement results data to the CSS DB system via ODBC.

Meter Data

Table Name: MV90_data_hour

Data Transfer Frequency: hourly at the top of the hour

The MDA system shall transfer 5-minute data to the PSS and thence to the Replica Database for access by the SPPC's CSS.

Price Curve

Table Name: Source_Price_Curve

Data Transfer Frequency: Daily

Standard format will be used to internally transfer price curve and time series data from Online SettlementTracker (Online PSS) to Online MDA for subsequent external data transfer to the Replica DB using the SPPC data model described in section 2.1.1.5 and import via the Offline MDA functionality before translation and export to the Offline SettlementTracker (Offline PSS).

Coefficient Data

Table Name: UDT_Coefficient_Data

Data Transfer Frequency: Yearly

Standard format will be used to internally transfer coefficient data from Online SettlementTracker (Online PSS) to Online MDA for subsequent external data transfer to the Replica DB using the SPPC data model described in section 2.1.1.5 and import via the Offline MDA functionality before translation and export to the Offline SettlementTracker (Offline PSS).

Availability Energy Notice

Table Name: UDT Availability_Energy_Notice

Data Transfer Frequency: Hourly

Standard format will be used to internally transfer availability energy notice data from Online SettlementTracker (Online PSS) to Online MDA for subsequent external data transfer to the Replica DB using the SPPC data model described in section 2.1.1.5 and import via the Offline MDA functionality before translation and export to the Offline SettlementTracker (Offline PSS).

Sunrise and sunset data

Table Name: UDT Sunrise_sunset

Data Transfer Frequency: Yearly

Standard format will be used to internally transfer sunrise and sunset data from Online SettlementTracker (Online PSS) to Online MDA for subsequent external data transfer to the Replica DB using the SPPC data model described in section 2.1.1.5 and import via the Offline MDA functionality before translation and export to the Offline SettlementTracker (Offline PSS).

Weather Data

Table Name: UDT Weather_data

Data Transfer Frequency: Daily

Standard format will be used to internally transfer weather data from Online SettlementTracker (Online PSS) to Online MDA for subsequent external data transfer to the Replica DB using the SPPC data model described in section 2.1.1.5 and import via the Offline MDA functionality before translation and export to the Offline SettlementTracker (Offline PSS).

Annual Projected Net Electrical Energy Target

Projected_UDT net_electrical_energy_target.

Data Transfer Frequency: Yearly

Standard format will be used to internally transfer projected NEE target data from Online SettlementTracker (Online PSS) to Online MDA for subsequent external data transfer to the Replica DB using the SPPC data model described in section 2.1.1.5 and import via the Offline MDA functionality before translation and export to the Offline SettlementTracker (Offline PSS).

Invoice Data

Table name: Calc_Invoice

Data Transfer Frequency: Monthly

Standard format will be used to internally transfer invoice data from Online SettlementTracker (Online PSS) to Online MDA for subsequent external data transfer to the Replica DB using the SPPC data model described in section 2.1.1.5 and import via the Offline MDA functionality before translation and export to the Offline SettlementTracker (Offline PSS).

Forecasted Data

Table Name: Forecasted_data.

Data Transfer Frequency: daily

Standard format will be used to internally transfer forecast data from Online SettlementTracker (Online PSS) to Online MDA for subsequent external data transfer to the Replica DB using the SPPC data model described in section 2.1.1.5 and import via the Offline MDA functionality before translation and export to the Offline SettlementTracker (Offline PSS).

Weather Data

Standard format will be used to internally transfer weather data from Online SettlementTracker (Online PSS) to Online MDA for subsequent external data transfer to the Replica DB using the SPPC data model described in section 2.1.1.5 and import via the Offline MDA functionality before translation and export to the Offline SettlementTracker (Offline PSS).

Table name: udt_weather_data

Data Transfer Frequency: Daily

Sun info

Standard format will be used to internally transfer suninfo data from Online SettlementTracker (Online PSS) to Online MDA for subsequent external data transfer to the Replica DB using the SPPC data model described in section 2.1.1.5 and import via the Offline MDA functionality before translation and export to the Offline SettlementTracker (Offline PSS).

Table Name: udt_sunrise_sunset

Data Transfer Frequency: Daily

The Online PSS shall be responsible for the pushing of both input and results data by ODBC to the Replica DB and thence to the Offline PSS. The Offline PSS shall then be responsible for the pulling of both input and results data by ODBC from the Replica DB.

4.1.4. Replica Database

The replica database at the SPPC site shall be hosted in a SQL Server database. The Replica database schema will comprise extracts of the SettlementTracker database table model to capture the export of User Defined Table data, Metered and SCADA data and Time Series data. The replica database shall receive this data from the Online SettlementTracker (PSS) database using ODBC communication across an encrypted transport protocol.

Both the Offline and Online Plant Settlement Systems shall have only “Insert” and “Read” abilities to the Replica database.

The data push from the Online SettlementTracker to the Replica database shall be triggered by the completion of the Monthly Settlement Calculations by the Online SettlementTracker database as a post process instruction.

The data pull from Replica Database to the Offline SettlementTracker database shall be a scheduled operation, to take place at a regular interval of 1-hour. The Offline SettlementTracker shall request the hours data from the Replica DB via ODBC protocol all data updated in respect of the UDT, Metered data, SCADA data and Time Series SettlementTracker tables.

The Replica Database shall also receive raw meter data and PV SCADA data from the MDA and Calculated Settlement Results from both the Online and Offline systems via ODBC protocol.

This data will be stored in two tables (a data table and a status table) for each data source.

The relationship between the status table and its related data table is one to many, depending upon the (related time – specific hour, specific monthly invoice), which will define it in the table structure under the column name [Group ID].

The mechanism operates as follows:

4.1.4.1. For Hourly data sources

All hour's tag information is inserted into the corresponding data table at the top of each hour as shown in the Table Structure.

[Group ID] as previously explained.

The value of the [Internal Seq. in Group ID] field will commence with 1 and increment by 1 for each new record added up to the maximum number of tags which relate to a specific hour or month.

When the data insert process is completed for a given hour's information a single value of 1 bit will be added to that data source's status table for the [Check Status] field, indicating the successful completion of the data insertion to the Replica DB.

4.1.4.2. For Monthly data sources

For monthly data sources it is a similar situation. All the month's tag information is inserted to the corresponding data table as shown in the Table Structure section below. When the data insert process is completed for the given month's and version's information a single record is inserted to the associated status table with the value of 1 bit in the [Check Status] field.

4.1.5. Replica DB Table Names and types

#	Table Name	Table Description	Table Type
1	<PLANT NAME>TagsSMD	<PLANT NAME> Settlement Meter Tag Data	Data
2	<PLANT NAME>TagsSMD_Status	<PLANT NAME> Settlement Meter Data tag update Status	Status
3	<PLANT NAME>TagsDCMS	<PLANT NAME> Distributed Control Management System (SCADA) Tag Data	Data
4	<PLANT NAME>TagsDCMS_Status	<PLANT NAME> Distributed Control Management System (SCADA) Tag update Status	Status
5	<PLANT NAME>TagsHonTCM	<PLANT NAME> Hourly Online Tariff Calculation Mechanism Tags data	Data
6	<PLANT NAME>TagsHonTCM_Status	<PLANT NAME> Hourly Online Tariff Calculation Mechanism update status	Status
7	<PLANT NAME>TagsHoffTCM	<PLANT NAME> Hourly Offline Tariff Calculation Mechanism Tag data	Data
8	<PLANT NAME>TagsHoffTCM_Status	<PLANT NAME> Hourly Offline Tariff Calculation Mechanism update status	Status
9	<PLANT NAME>TagsHonPPM	<PLANT NAME> Hourly Online Plant Performance Monitor Tag data	Data
10	<PLANT NAME>TagsHonPPM_Status	<PLANT NAME> Hourly Online Plant Performance Monitor update status	Status
11	<PLANT NAME>TagsHoffPPM	<PLANT NAME> Hourly Offline Plant Performance Monitor Tag data	Data
12	<PLANT NAME>TagsHOffPPM_Status	<PLANT NAME> Hourly Offline Plant Performance Monitor update status	Status
13	<PLANT NAME>TagsMOnTCM	<PLANT NAME> Monthly Online Tariff Calculation Mechanism Tag Data	Data

#	Table Name	Table Description	Table Type
14	<PLANT NAME>TagsMOnTCM_Status	<PLANT NAME> Monthly Online Tariff Calculation Mechanism update status	Status
15	<PLANT NAME>TagsMOffTCM	<PLANT NAME> Monthly Offline Tariff Calculation Mechanism Tag Data	Data
16	<PLANT NAME>TagsMOffTCM_Status	<PLANT NAME> Monthly Offline Tariff Calculation Mechanism update status	Status
17	<PLANT NAME>TagsMOnPPM	<PLANT NAME> Monthly Online Plant Performance Monitor tag data	Data
18	<PLANT NAME>TagsMOnPPM_Status	<PLANT NAME> Monthly Online Plant Performance Monitor update status	Status
19	<PLANT NAME>TagsMOffPPM	<PLANT NAME> Monthly Offline Plant Performance Monitor Tag Data	Data
20	<PLANT NAME>TagsMOffPPM_Status	<PLANT NAME> Monthly Offline Plant Performance Monitor update status	Status

Table 1: Replica DB Raw data & Settlement Results Table Structure

4.1.5.1. Table Structure

Data Table Definition

#	Field Name	Field Description
1	[PP Name]	This field represents the Data Source refer to **Error! Reference source not found.** below for details
2	[Tag Name]	This field represents the tag name from the source system
3	[Tag Time Stamp]	This is the time at which the tag data has been measured for hourly sources. For monthly sources, if the tags represent hourly sources, it will be same case as for hourly source. However, if the tags are not representing hourly values (e.g., contract values, version, date etc), it should be linked to time stamp of the given month which linked to the invoice (YYYY/MM/DD HH:MM, 20xx/xx/xx 00:00) and not to modified/created/activated timestamp. The time should be the top of each hour.
4	[Tag Value]	The measurement value of the tag at the given time (linked to given [Tag Time Stamp]). The maximum precision is 5 decimal places.
5	[Access Time Stamp]	This is the time when the information of the tag has been inserted into the Replica DB. In other words, it is when the insert query has been executed. The value of it in data tables usually is not the same as in Status Tables as there may be a small gap in time between the execution of the two insert processes
6	[Tag Quality]	This value represents the quality of [Tag Value] according to the measurement instrument. It is always equal to 1 or 0 not NULL. 1 means good quality and 0 means bad quality. TCM and PPM data tags shall always be given a value of 1

#	Field Name	Field Description
7	[Group ID]	This shows the one-to-many relationship which is supposed to be available between the Status table and its associated data table. It is a unique value (in the Status Table only) linked to all values of a given hour in the hourly sources (tables) or given month in the monthly sources (tables). It's start value is 1 and it shall always increment by 1. This gives the data of each hour a unique value for hourly sources as it is grouping the whole hour information by giving this hour a unique value when pushed/inserted together to represent one single hour. However, this is not the case with monthly data sources. Grouping all tags given for a specific month's information, which when pushed/inserted together represent one specific version of monthly data.
8	[Internal Seq. in Group ID]	Represents the sequence of tags which relate to certain Group ID hour/month

Table 2: Data Table Definition

Status Table Definition

#	Field Name	Field Description
1	[Group ID]	This shows the one-to-many relationship which is supposed to be available between the Status table and its associated data table. It is a unique value (in the Status Table only) linked to all values of a given hour in the hourly sources (tables) or given month in the monthly sources (tables). It's start value is 1 and it shall always increment by 1. This gives the data of each hour a unique value for hourly sources as it is grouping the whole hour information by giving this hour a unique value when pushed/inserted together to represent one single hour. However, this is not the case with monthly data sources. Grouping all tags given for a specific month's information, which when pushed/inserted together represent one specific version of monthly data.
2	[Internal Seq. in Group ID]	Represents the sequence of tags which relate to certain Group ID hour/month
3	[PP Name]	This field represents the Data Source refer to **Error! Reference source not found.** below for details
4	[Access Time Stamp]	This is the time when the information of the tag has been inserted into the Replica DB. In other words, it is when the insert query has been executed. The value of it in data tables usually is not the same as in Status Tables as there may be a small gap in time between the execution of the two insert processes
5	[Check Status]	This is an indicator of completion of the insertion SQL statement in the data table. It has only two values and not including NULL. Either one (1) which means completed or zero (0) which means not completed. Usually the value is one (1) in most cases. However, there are a few abnormal cases where the value is zero (0). The method has been explained in 4.1.4.

Table 3: Status Table Definition

4.1.6. SCADA Data signals required for PSS & SPM

The following table illustrates the PV SCADA data signals required by PSS and SPM to perform settlement and performance ratio calculations. These data signals will be provided with a granularity of 1-hour. The signal values will be either summated across the hour or averaged for the hour as indicated in the following table. The data signals will be shared from the Message Transfer system via the ODBC interface.

#	Title	CATEGORY
1	METEOROLOGICAL STATION AMBIENT AIR TEMPERATURE	METEOROLOGICAL STATION
2	METEROLOGICAL STATION AMBIENT RELATIVE HUMIDITY	METEOROLOGICAL STATION

#	Title	CATEGORY
3	METEOROGICAL STATION WIND SPEED	METEOROLOGICAL STATION
4	METEOROGICAL STATION WIND DIRECTION	METEOROLOGICAL STATION
5	METEOROLOGICAL STATION IN-PLANE IRRADIANCE	METEOROLOGICAL STATION
6	METEOROLOGICAL STATION REAR SIDE IRRADIANCE	METEOROLOGICAL STATION
7	METEOROLOGICAL STATION HORIZONTAL UP IRRADIANCE	METEOROLOGICAL STATION
8	METEOROLOGICAL STATION HORIZONTAL DOWN IRRADIANCE	METEOROLOGICAL STATION
9	METEOROLOGICAL STATION HORIZONTAL ALBEDO	METEOROLOGICAL STATION
10	METEOROLOGICAL STATION PV MODULE TEMPERATURE 1	METEOROLOGICAL STATION
11	METEOROLOGICAL STATION PV MODULE TEMPERATURE 2	METEOROLOGICAL STATION
12	METEOROLOGICAL STATION PV MODULE TEMPERATURE 3	METEOROLOGICAL STATION
13	METEOROLOGICAL STATION PV MODULE TEMPERATURE 4	METEOROLOGICAL STATION
14	METEOROLOGICAL STATION PV MODULE TEMPERATURE 5	METEOROLOGICAL STATION
15	METEOROLOGICAL STATION PV MODULE TEMPERATURE 6	METEOROLOGICAL STATION
16	METEOROLOGICAL STATION DAILY CLEAN PV MODULE SOILING RATIO UP	METEOROLOGICAL STATION
17	METEOROLOGICAL STATION DAILY CLEAN PV MODULE SOILING RATIO DOWN	METEOROLOGICAL STATION
18	METEOROLOGICAL STATION NON-CLEAN PV MODULE SOILING RATIO UP	METEOROLOGICAL STATION
19	METEOROLOGICAL STATION NON-CLEAN PV MODULE SOILING RATIO DOWN	METEOROLOGICAL STATION
20	TOTAL ACTIVE POWER OUTPUT OF ALL INVERTERS IN BLOCK #NN	DATALOGGER

#	Title	CATEGORY
21	TOTAL POWER FACTOR OF ALL INVERTERS in BLOCK #NN	DATALOGGER
22	TOTAL REACTIVE POWER OUTPUT OF ALL INVERTERS in BLOCK #NN	DATALOGGER
23	CURTAILMENT INSTRUCTION	PPC
24	WEATHER FORECAST TEMPERATURE	WEATHER FORECAST
25	WEATHER FORECAST DEWPOINT	WEATHER FORECAST
26	WEATHER FORECAST CLOUD COVER %	WEATHER FORECAST
27	WEATHER FORECAST HUMIDITY	WEATHER FORECAST
28	WEATHER FORECAST WIND SPEED	WEATHER FORECAST
29	WEATHER FORECAST WIND DIRECTION	WEATHER FORECAST
30	WEATHER FORECAST PRECIPITATION	WEATHER FORECAST
31	WEATHER FORECAST WET BULB	WEATHER FORECAST

Table 4: SCADA Data Signals required for PSS & SPM

The hourly SCADA signals identified in table 1 will be shared by the MDA with both Online and Offline systems but other raw SCADA signals will not be routed through the Online and Offline PSS/SPM systems.

4.1.6.1. Meter Data required for PSS/SPM

The meter data required for settlement calculations in the Online and Offline systems is the ‘Meter Active Power (kWh)’ from the settlement meters located near the point of interconnection with the transmission grid.

This active power meter data is sourced from the MDA and will in turn be sourced from either the primary or backup meter depending upon which meter is the ‘good’ data source during the relevant settlement period.

4.1.7. Dispatch Signals

In order to calculate the Deemed Net Electrical Energy in accordance with clauses 7.7.2, 8.3, 8.4, 14.7, 14.8 and Schedule 17 of the PPA the SettlementTracker shall require the notification of each hourly interval where (A) a dispatch instruction requires the IPP to limit or reduce the Net Electrical Energy produced by the Plant, (B) a failure of the electricity transmission facilities prevents/delays completion of acceptance testing, (C) the effects of Force Majeure either before or after PCOD.

The NGSA’s DLC system operator telephones the Arrass Renewable Energy Company’s plant operations team to inform them of the plant’s production curtailment. The instruction may be for immediate curtailment or after some time (e.g., 15 min / 1hour). The Arrass Renewable Energy Company’s station is operated with a Power Plant Controller (PPC), used to regulate, and control the networked inverters, devices and equipment at the Al-K solar PV plant, in order to meet specified setpoints. The plant operations team enters the received curtailment instructions start time and quantity into the PPC. The instruction is then immediately implemented by the PPC.

4.1.8. Data Capture

4.1.8.1. By User Interface

The following data items shall be manually captured either through the graphical user interface (GUI) of the SettlementTracker application or via an MS Excel template file import, in the event of the invocation of a Force Majeure or Offtakers Risk Event:

- The start date of a counterparty's invocation of Force Majeure or an Offtaker's Risk Event in accordance with the provisions of their PPA clause 14.
- The cessation date of a counterparty's invocation of Force Majeure or an Offtaker's Risk Event in accordance with the provisions of their PPA clause 14.

4.1.8.2. By manual file upload

The following data items shall be manually captured in an MS Excel file template and then the Contract Analyst shall manually initiate the file import to SettlementTracker' s user defined tables.

4.1.8.3. Semi-Static Data: PPA Contract Parameters

Acronym	Title	Value	UoM
CCR	Capital Cost Recovery component of Energy Charge	5.22895	Hals/kWh
Yccr	Foreign portion of capital cost recovery component	100.00	%
OMR	Operations and Maintenance Cost Recovery component of Energy Charge	0.91309	Hals/kWh
Yomr	Foreign portion of O&M cost recovery component	40.00	%
P_{mod}	the sum of the nameplate rating of the PV modules at STC conditions	300	MW
Gstc	Irradiance at standard temperature conditions	1	kW/m^2
β	the absolute (unsigned) value of the temperature coefficient from the Solar PV module's data sheet for the module's nominal power output.	0.34%/°C	%/℃
$Tmod_m$	Projected Monthly Average Module Temperature set forth in sheet D of Schedule 1	Refer to Table 18	%
ICOD	Initial Commercial Operation Date	30/06/2022	DateTime
PCOD	Project Commercial Operation Date	31/12/2022	DateTime
PNOM Proposal	Project Nominal Installed Power Rating Proposal[1]	300	MW_{AC}
PNOM Required	Project Nominal Installed Power Rating required[6]	>=285	MW_{AC}
ICT	Industrial Consumption Tariff	18[2]	Hals/kWh

Table 5: PPA Semi-Static Contract Parameters

4.1.8.4. Reference Values for Inflation indices, Interest Rates and Exchange Rates

Acronym	Title	Value	Reference Date
USPPI	United States of America Producer Price Index	194.3	01/01/2019
KSACPI	Kingdom of Saudi Arabia Consumer Price Index[3]	105.1	01/01/2019
EXR	Exchange Rate USD: SAR	3.75 USD	01/01/2019

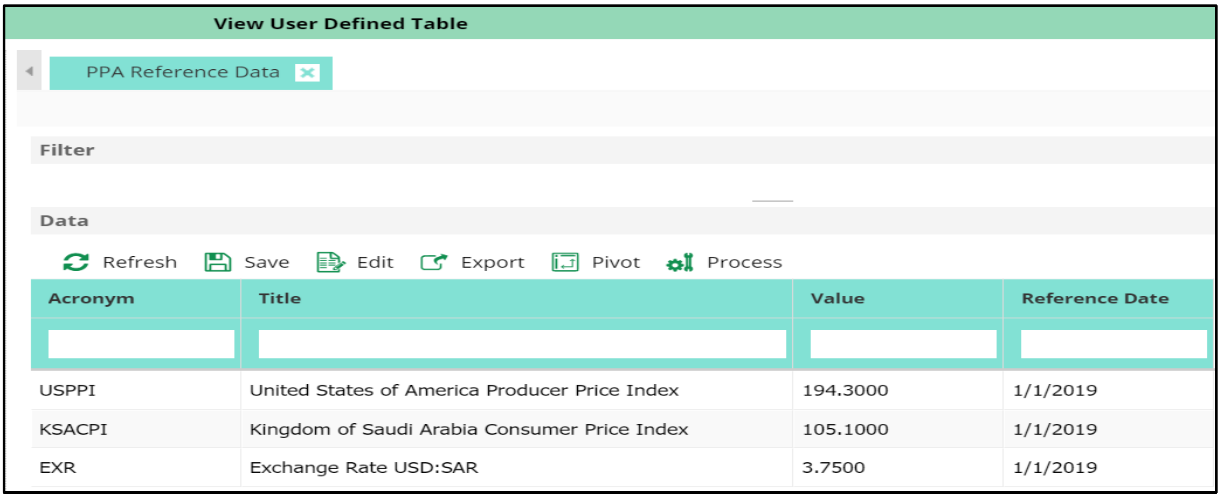
Table 6: Reference Values from Attachment B to Schedule 9 Indexation/Adjustment of Energy Charge

Required data for PPA reference will be captured in the User defined table in SettlementTracker as illustrated below in Figure 5.

[1] Refer to file Appendix_12 Performance_Guarantee_Data_rev1.pdf for Power Guarantees at PCOD.

[2] Industrial Consumption Tariff rate ssource: Http://se/com/sa/en-us/Customers/Pages/TariffRates.aspx

[3] Assumption: That the KSACPI values to be used as the reference index shall be that of the "Housing, Water, Electricity, Gas and Other Fuels" division.



View User Defined Table

PPA Reference Data

Filter

Data

Refresh Save Edit Export Pivot Process

Acronym	Title	Value	Reference Date
USPPI	United States of America Producer Price Index	194.3000	1/1/2019
KSACPI	Kingdom of Saudi Arabia Consumer Price Index	105.1000	1/1/2019
EXR	Exchange Rate USD:SAR	3.7500	1/1/2019

Figure 5: User Defined Table

4.1.9. Time Series Data Capture: Inflation Indices, Interest Rates and Exchange Rates

The following data items shall be manually captured in an MS Excel file template each day and then immediately following the end of the month the Contract Analyst shall manually initiate the file import to SettlementTracker's user defined tables.

- SOFR (Secured Overnight Financing Rate) (to be confirmed following PPA Availability)
- SAIBOR (Saudi Arabian Interbank Offer Rate for three-month SAR deposits) published on the Reuters-Thompson screen page "SUAA" on the day immediately succeeding the due date for the relevant payment. (To be confirmed following PPA Availability)
- EXR (USD: SAR Exchange Rate) quoted by SAMA for each relevant day.

These timeseries values differ from the Reference Values in that they are continually updated daily/monthly throughout the duration of the PPA as relevant and are for use only with respect to specific Billing Period while the reference values are singular values as at a specific point in time for repeated use during Billing Periods across the lifetime of the PPA.

4.2. Plant Availability – Energy Notice

The following data items shall be manually captured in an MS Excel file template each day and then immediately following the end of the month the Contract Analyst shall manually initiate the file import to SettlementTracker's user defined tables.

- Day-Ahead availability -Energy Notice, prepared and issued by the plant operator to the Offtaker by 11:00am on the day prior to planned dispatch.
- Superseding Availability Energy Notice, prepared and issued by the plant operator to the Offtaker as soon as it is known that a previously issued Availability-Energy Notice(s) must be updated.

For the avoidance of doubt the PSS/SPM system shall import, store and report the Plant's availability-energy notices but does not determine the availability for the energy-notice, nor is it responsible for notifying the Offtaker of the energy notices.

4.3. User/ Roles and Privileges

Users can be configured as per license during project implementation. The following contains standard roles and privileges. The roles in Settlement Tracker are configurable. These roles will be revised as necessary during implementation.

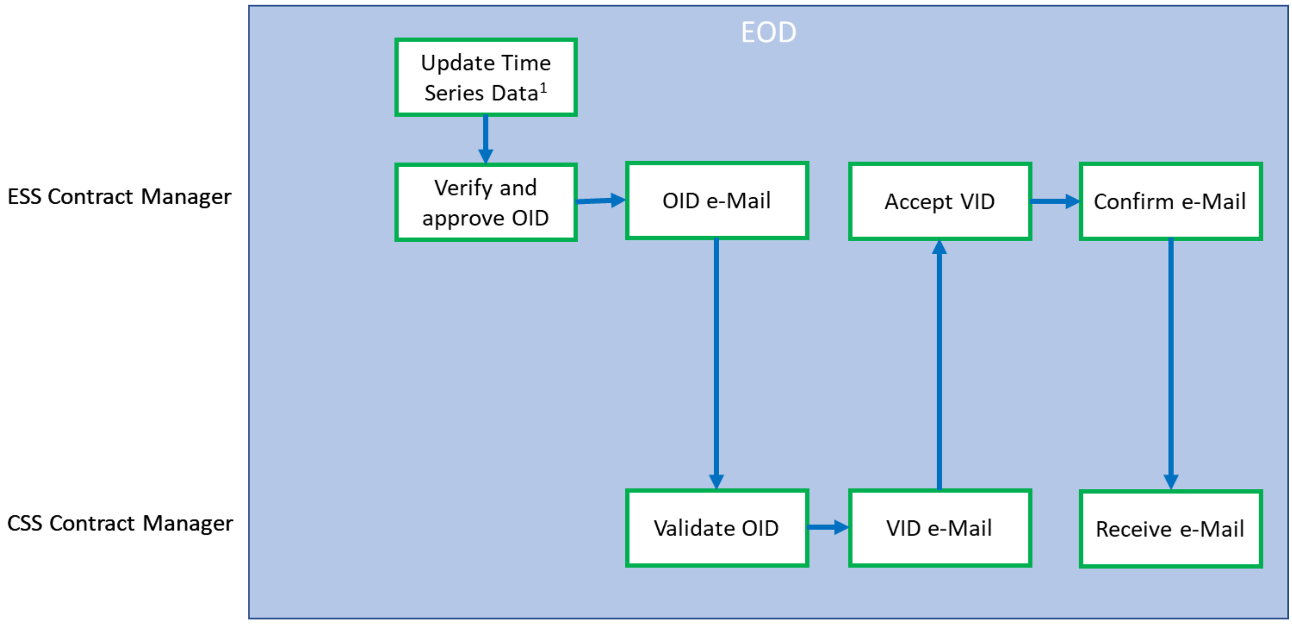
- Contract Manager
- Contract Analyst
- Administrator

4.3.1. Data Migration

The Arrass Solar PV plant PSS is a greenfield implementation and therefore there is no requirement for the migration of data from previously operated software systems.

4.3.2. End of Period Processes

This section discusses requirements end of Day (EOD).



1. Time series data includes, MDA metered data, DCS sensor data, inflation indices, interest rates and exchange rates

Figure 6 Arrass PSS End of Day Process

The Original Input Data (OID)[4] report shall comprise only the determinants necessary to calculate the settlement invoice(s) for the Billing Period (a calendar month) including:[5]

- IPNOM – Initial Nominal Installed Power Rating
- PNOM – Nominal Installed Power Rating

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[5] ASSUMPTION 5: The Original Input Data shall include only that data necessary to calculate the settlement invoices. The validation of plant operational and planning data shall be managed by IPP.

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- CCR – Capital Cost Recovery component
- OMR – Operations and Maintenance cost recovery component
- Y_{ccr} – Foreign proportion of capital cost recovery component
- Y_{omr} – Foreign proportion of operation and maintenance cost recovery component
- SAIBOR – Saudi Arabia Interbank Offer Rate for each day of the relevant month
- SOFR (Secured Overnight Finance Rate) for each day of the relevant month.
- EXR – USD to SAR exchange rate for each day of the relevant month
- USPPI – United States Producer Price Index for the relevant month
- KSACPI – Kingdom of Saudi Arabia Consumer Price Index for the relevant month
- Deemed Energy $Flag_{(i)}$ for each interval 'I' (hour) of the Billing Period
- Value Added Tax (VAT) $rate_n$ (%) applicable.
- $NEE_{i, actual}$ (hourly metered data for both exported Plant generation and imported Plant consumption for the duration of the Billing Period)
- $GPOA_i$ (SCADA PV data from pyranometer sensors of solar irradiation at the plane of array) for each interval 'i' (hour) of the billing period
- Coefficients A, B and $C_{y, n}$ for the relevant Contract Year and month.
- Industrial Consumption Tariff rate (Hals/kWh) applicable

S.N	Title	Description
1	Update Time Series Data	At the end of each day both the Online SettlementTracker and the Offline SettlementTracker shall update all Timeseries data [6](including meter data, inflation indices, interest rates and exchange rates) from the daily Time Series import spreadsheet saved to the appropriate directory by the Contract Analyst.
2	Verify and Approve Original Input Data	SettlementTracker shall verify that for all data items included within the OID daily set that the values are: i. The correct data type (e.g., integer, float, string) ii. Range a number between permitted range where specified. iii. Unique where required e.g., record identifier code. iv. No null values in mandatory fields Note: On the Business Day following a Non-Business Day the OID set for verification shall include the preceding Business Day and all contiguous following Non-Business Days. The Contract Analyst and Contract Manager shall be notified by SettlementTracker automatically if data validation should result in an error e.g., unexpected null values.
3	OID e-Mail	In the event that the SettlementTracker shall successfully verify the OID a pro-forma e-mail shall be produced and sent to the counterparty contract manager's email address with the OID report file attached.
4.	Validate OID	Before the end of the next business day following receipt of the OID data, The SPPC Counterparty shall compare the OID report received from IPP to the Shadow OID

[6] ASSUMPTION 6: The Online shall update their timeseries data records for interest rates and exchange rates on a daily basis.

ASSUMPTION 7: The Online shall update their timeseries data records for inflation indices on a monthly basis.

S.N	Title	Description
		report produced by the Offline SettlementTracker and identify any discrepancies.
5	VID e-Mail	By the end of next business day following receipt of the OID data shall communicate by email to IPP a report highlighting any deviations between their results. If no VID is submitted by the SPPC to IPP then IPPI proceed at the month end to generate invoices using the OID.
6	Accept VID	IPP shall review the discrepancy report and if none are reported shall confirm its acceptance of the validated input data from the SPPC before the end of the next business day after receipt of the Validated Input Data (discrepancy) report.
7	Confirm e-Mail	By the end of the next business day after receipt of the Validated Input Data (discrepancy) report the IPP shall e-Mail a confirmation to the SPPC of its agreement with the Validated Input Data if there are no discrepancies between the OID and VID.
8	Receive e-Mail	The SPPC shall receive the confirmation of VID from the IPP. In the case there is a dispute between SRRECIPPSPPC the two parties shall attempt to resolve the dispute within two Business Days

Table 7: End of Day Workflow

The Offtaker upon receipt of the OID report shall validate the data by comparison to their shadow OID report produced by the Offline SettlementTracker and reply by email either confirming the data is matched and validated input data (VID) or identifying the disputed values.

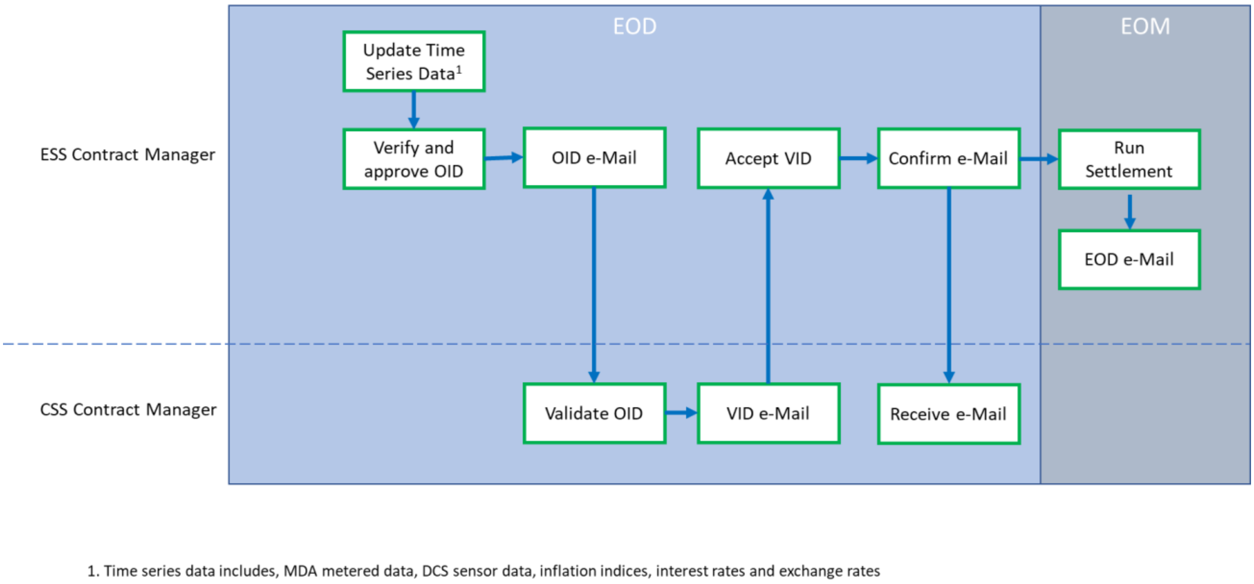


Figure 7: End of Month Process

4.3.3. Valuation Process

Upon settlement process, SettlementTracker calculates the charges (by evaluating the underlying formula setup) and presents the results for each charge in the settlement GUI and reports.

4.3.4. Project Delivery Approach

The project team will configure the SettlementTracker application to deliver the PSS functionality described within this Implementation Specification (ISPEC) document through a series of consecutive two-week duration sprints. Each sprint being the set period of time during which specific configuration tasks will be completed and made ready for system testing.

The project team shall prepare a Factory Acceptance Testing (FAT) procedure document which will be describe the approach to performing Factory Acceptance Tests and will be delivered to GPEC for review and approval. GPEC shall ensure that IPP and SPPC review and provide their approval of the FAT procedure document within one week of receipt.

Following approval of the FAT Procedure document the FAT shall be performed in accordance with the documented procedures at the Hitachi Energy site. Documentary evidence of the FAT performed shall be made available to GPEC and GPEC shall ensure that IPP and SPPC shall review and provide their approval of the FAT within one week of receipt.

The Project team shall prepare a User Acceptance Testing (UAT) or system testing procedure document which will describe the scope of UAT and type of tests to be performed (e.g., Integration, functional/system testing and also those tests excluded e.g., existing product functionality and repetition of system functionality between the two instances of online and offline PSS. This will be delivered to GPEC for review and approval. GPEC shall ensure that IPP and SPPC review and provide their approval of the UAT procedure document.

4.4. Deployment Architecture

There shall be two instances of the Plant Settlement System (PSS) deployed in compliance with the SPPC's Interface Specification v5.0.

1. The online instance of the PSS shall be deployed at the IPP power station site in single panel containing the primary and backup MDA systems.
2. The offline instance of the PSS will be deployed at the SPPC datacenter in a single Panel. This shall comprise the PSS application and DB server.

4.4.1. Data Management

Settlement Tracker Database hosts many kinds of data and managing this data is a vital process as a part of the maintenance of the application. Archiving is the process of moving data from the production database to an archival database. Archiving helps to keep the database size smaller and more manageable. It will also help in maintaining system performance goals.

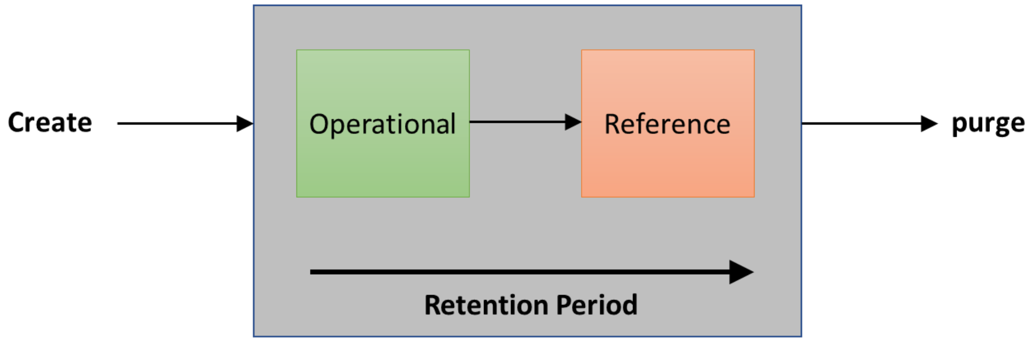


Figure 8: Data Management Process

The following data will be archived as a part of archival process in Settlement Tracker. Archival policy will be further defined during the project.

- Price Curve (calculated/derived curves from price curves)
- Meter data
- Forecasted Data

Module	Price Curve
Partition/Archival Field	as_of_date maturity_date
Number of data in main table	Last two years data for settlement price
Archival Frequency	Yearly
Purge filter	Price_curve as_of_date < Today – 2 years.
Purge Filter	What data can be purged?

Figure 9: Price Curve

Module	Meter Data
Partition/Archival Field	Term start
Number of data in main table	Last two years data
Archival Frequency	Yearly
Purge filter	Term start < Today – 2 years.
Purge Filter	What data can be purged?
Purge Filter	What data can be purged?

Figure 10: Meter Data

Module	Forecasted Data
Partition/Archival Field	Term Start
Number of data in main table	Last five years
Archival Frequency	Yearly
Purge filter	Term Start < Today – 5 years
Purge Filter	What data can be purged?

Figure 11: Forecasted Data

4.5. Data Structure

4.5.1. Data Management

The following diagram is a representation of the business process for operating SettlementTracker. The user shall be required to perform these business processes for the proper functioning of the system.

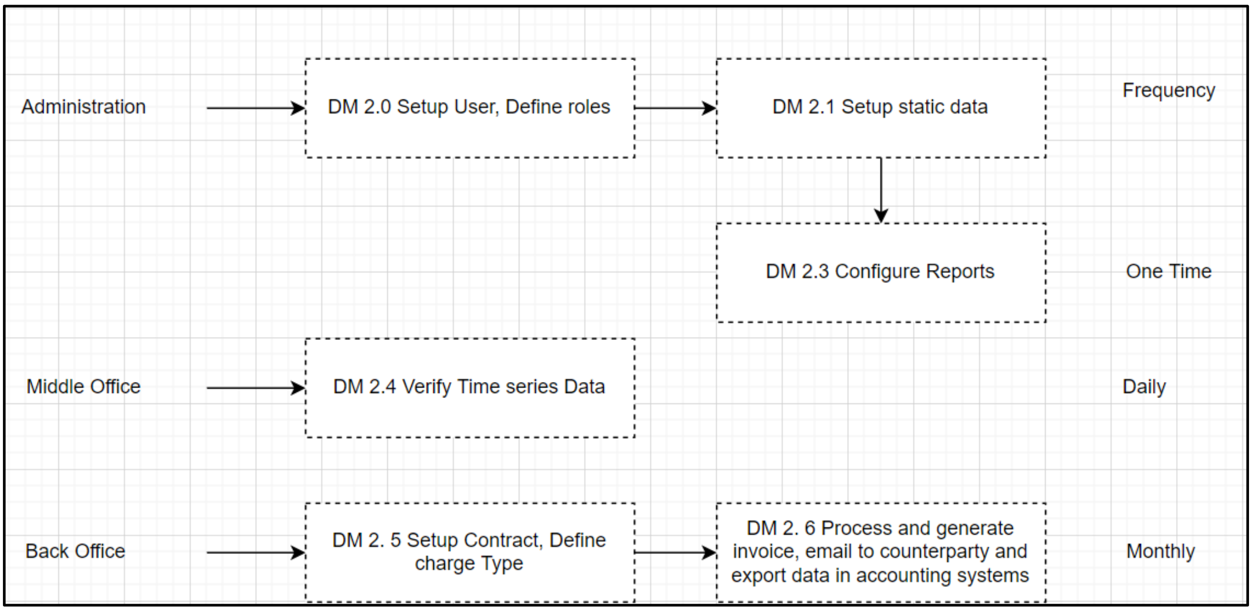


Figure 12: Data Management

4.5.1.1. Administration

- Hitachi Energy Team will setup users during the implementation of a project and configure standard roles.
- Setup Static data DM 2.1

Data type	Definition
Commodity	Commodity Definition
Expiration Calendar	Expiration (settlement or publication) date for products
Holidays	Lists of different holidays
Location	Locations name and relevant details add interconnect (naming conventions – path, interconnect, etc.)
Price Curve Definition	Price Curve Definition
Traders	Traders' definition
UOM	Unit of Measures
Meter	Meter Name and relevant detail

Table 8: Master & Static Data to be configured.

Standard and custom reports will be configured DM 2.3

4.5.1.2. Middle Office

Required timeseries data should be uploaded and verified manually DM 2.4 consistent with the availability of the published timeseries data i.e daily or monthly as appropriate to the data.

4.5.1.3. Back Office

- Setup contract, mapped required charges for invoicing, map invoice template, configure email for emailing, define payment date DM 2.5
- Process invoice, verify calculation, generate document and email to counterparty DM 2. 6

4.5.2. Solar Performance Monitoring

The Solar Performance Monitoring functional requirements include the following:

1. Calculate the Actual Performance Ratio (APR) for a user-specified period of measurement intervals 'i' (hours) This will initially be for the Performance Test to determine whether the plant has met the minimum acceptance criteria, but subsequently to monitor the degradation of the Plants performance over the contract life.
2. Calculate the Projected Performance Ratio (PPR) for a user specified period of measurement intervals 'i' (hours).
3. Determine the Deemed Energy Coefficients for each Contract Year, refer to Table 15 Items 20, 21 and 22 for the required formulae.

4.5.2.1. Actual Performance Ratio

The Actual Performance Ratio shall be calculated as follows:
Unlike the Settlement calculations which shall be initiated following the end of each calendar month, for the APR calculation, the user must specify both the start date and time as well as the end date and time for which the calculation shall be performed before initiating the calculation the results of which will populate the Actual Performance Ratio report.
The Actual Performance Ratio shall be determined in accordance with the following formulae within the SettlementTracker Charge definition.

$$APR = \frac{\sum_{i=1}^{j} NEEact_i}{P_{mod}} * \frac{G_{stc}}{\sum_{i=1}^{i}\left(GPOA_i\left(1+\beta*(Tmod_m - Tmeas_i)\right)\right)}$$

Equation 1: Actual Performance Ratio

Where:

Acronym		Description
$NEEact_i$	=	The Net Electrical Energy metered at the connection point of Arrass Solar PV Plant to the NGSA transmission grid (measured in kWh over each metering interval (hour). Note this value may be either positive or negative as NEE value may be imports or exports across the connection. Although normally the calculation would be scheduled for a period when the plant's output has not been curtailed.
i	=	Index of one-hour metering interval

Acronym		Description
j	=	The number of hours within the user specified testing period
β[7]	=	The absolute (unsigned) value of the temperature coefficient from the Solar PV module's data sheet for the module's nominal power output (in %/℃)
$Tmod_m$[8]	=	Projected Monthly Average Module Temperature (in ℃) as set forth in Form Sheet D of Schedule 1 of the PPA
$Tmeas_i$	=	The actual average module temperature measured during each Metering Interval 'i' by the temperature sensors placed on the reverse side of the modules (in ℃) from PV FIELD IO list.xlsx "METEOROLOGICAL STATION PV MODULE TEMPERATURE #"
P_{mod}	=	The sum of the nameplate rating of the PV modules at Standard Temperature Conditions
$GPOA_i$	=	The irradiation in kWh/m^2 measured per each metering interval 'i' with an on-site pyranometer with an identical inclination as the PV modules; and from PV FIELD IO list.xlsx
Gstc	=	Irradiance at Standard Temperature Conditions (STC) (1kW/m^2)

Table 9: Actual Performance Ratio Determinants

The Actual Performance Ratio calculated shall be compared to the Guaranteed Performance Ratio (GPR) at PCOD for the relevant calendar month during which the Acceptance Tests and APR calculation are conducted.[9] Note that following the successful completion of Acceptance Testing at PCOD the Actual Performance Ratio will be expected to progressively decline in subsequent contract years and may therefore decline below the GPR during the twenty-five-year term of the PPA.

The Projected Performance Ratio is calculated using the same logic as APR but with the generation forecast data substituted for the outturn metered Net Electrical Energy quantity (NEEi, act) data.

The APR is determined by selection of a number of days that meet the commissioning criteria, while the PPR date range is restricted to the day-ahead forecast range.

4.5.2.2. Data set filtering pre-determination of Deemed Energy Coefficients

The determination of Deemed Energy Coefficients shall use the hourly metered values of Net Electrical Energy ($NEE_{i,\ actual}$) received from the MDA system and the hourly average Global Solar Irradiation on the Plane of Array ($GPOA_i$) values received from the PV SCADA system in respect of the Arrass Solar PV Plant. Any $GPOA_i$ and NEE_i for hours where either Force Majeure dates have been captured within the PSS or the Deemed Net Electrical Energy Notification report flag is set as one (1) shall be excluded from the calculation of the coefficients.

4.5.2.3. Determination of Deemed Energy Coefficients

The following formulae using Newton polynomials enable the determination of the deemed energy coefficients for A, B and C for a specific month 'n' of a specified contract year 'y' shall be performed using the following process.

1. The provided Excel Template document shall be created to determine the Coefficients for the quadratic equation using the MS Excel function 'LINEST'.
2. A configured 'Coefficient View' report shall return only the hourly values of GPOAi, $GPOA_i^2$ and positive values of NEEi, act (i.e., exported metered generation values only) for the user specified year. Note, deemed energy coefficients are calculated using the valid (unconstrained or force majeure declared) hourly values for the calendar year 'y-1' within 7 days of the end of a contract year 'y-1' of operation for use in the just started contract year 'y'.

SettlementTracker shall create the Coefficients Template document from the Menu➔Administration➔Reporting➔Excel Add-in➔Excel Document Generation menu and populate the template with the values returned by the Coefficient View for the Contract YearContract Yearend shall be determined as follows:

1. If the project Commercial Operation Date occurs on or before the 15th day of a calendar month then such calendar month shall be considered in its entirety as the first Billing Period of Contract Year 1 and subsequent Contract Years, i.e., that the first Billing Period shall be 1st December ending on the last billing period on 31st November.

2. If the project Commercial Operation Date occurs on or after the 15th day of a calendar month then such calendar month is considered in its entirety as the last Billing Period of Contract Year 1 and subsequently any following Contract Years, i.e., that the first Billing Period shall start 1st January ending on the last Billing Period on 31st December.

SettlementTracker will export a report of the hourly values of GPOA and NEE for a period of a year, to a Microsoft Excel template and using the LINEST function to determine the coefficients for the curve which describes the relationship between GPOA and NEE.

Coefficient $A_{y,n}$

$$A_{y,n} = \frac{\frac{NEE_{i+2} - NEE_{i+1}}{GPOA_{i+2} - GPOA_{i+1}} - \frac{NEE_{i+1} - NEE_{i}}{GPOA_{i+1} - GPOA_{i}}}{GPOA_{i+2} - GPOA_{i}}$$

Equation 2: Coefficient A

$$B_{y,n} = \frac{NEE_{i+1} - NEE_{i}}{GPOA_{i+1} - GPOA_{i}}$$

Coefficient $B_{y,n}$

Equation 3: Coefficient B

Coefficient $C_{y.n}$

$$c_{y,n} = GPOA_{i}$$

Equation 5: Coefficient C

Where:

Acronym		Description
$A_{y,n}$	=	The deemed energy coefficient A for contract year ‘y’ and month ‘n’
$B_{y,n}$	=	The deemed energy coefficient B for contract year ‘y’ and month ‘n’
$C_{y,n}$	=	The deemed energy coefficient C (also constant) for contract year ‘y’ and month ‘n’
NEE_i	=	The Net Electrical Energy metered for interval ‘i’ (hour) at the connection point of Arrass Solar PV Plant to the NGSA transmission grid (measured in kWh over each metering interval (hour).
NEE_{i+1}	=	The Net Electrical Energy metered at the connection point for interval ‘i’ +1-hour e.g. where i = 10:00 then i+1 = 11:00
NEE_{i+2}	=	The Net Electrical Energy metered at the connection point for interval ‘i’ +2-hour e.g. where i= 10:00 then i+2 = 12:00
$GPOA_i$	=	The irradiation in kWh/m^2 measured per each metering interval ‘i’ with an on-site pyranometer with an identical inclination as the PV modules

Table 10: Deemed Energy Coefficient determinants

4.5.3. Generation Forecast

The Forecast module application will use flexible neural network, weather adaptive program forecasting to produce an hourly solar production forecast. Neural Network models provide the advantage of generating forecasts by learning from actual historical data, they recognize complex relationships and patterns in the data, resulting in more accurate modelling of non-linear relationships.

Initially the short-term forecasting neural network model employed by IPP at the Online PSS will be trained using GPEC historical weather data for the Arrass site and GPEC projected generation data. Following go-live the forecaster may be periodically further trained as additional historical weather data and actual generation data for the plant site becomes available.

The model will use weather data, solar data and meter data to produce hourly forecasts for the remainder of the day (intraday), for the next day (day ahead) and for the hourly deemed net electrical energy for next 24 hours upon receipt of a curtailment instruction. This forecast will run once every 30 minutes automatically.

Static inputs into the neural network model will include:

- Threshold
- Maximum step
- Learning rate
- Repetition
- Hidden layers
- Algorithm
- Error function

Hitachi Energy

Dynamic inputs into the neural network model will include the following hourly data:

- Historic/Actual generation output
- Temperature (forecast)
- Cloud cover (forecast)
- Precipitation (forecast)
- Humidity (forecast)
- Dewpoint (forecast)
- Sunrise time
- Sunset time
- Solar production during the day

The forecast weather data shall be provided by PV SCADA to the MDA system and then imported into SettlementTracker through the ODBC interface.

The neural network model will output the data into the application Forecast profiles which can be used for various things elsewhere such as settlement calculations. The application will include a chart where actual and forecasted values for each model can be compared.

The forecaster module at IPP's online instance of the PSS shall be trained to produce a forecast of generation based upon the declared plant availability notice, forecast weather data and history of outturn weather data for the site. This forecast shall be made available to the SPPC's offline instance of the PSS in the same manner as other time series data. The SPPC's offline PSS shall not be capable of reproducing the forecast generation from the online instance of the PSS. This is because the training of the neural network with historical data at two distinct instances of the software cannot guarantee that the resulting forecast will be identical. It is a feature of neural network training that while the system can identify and take account of non-linear relationships in the data and demonstrate accuracy of forecasting over time. It is not possible to ensure the identical training data will result in identical relationships being identified and the resulting forecasts being perfectly identical.

4.5.4. Plant and Accounting Settlement System (Back Office)

Process and view invoice UI will be used and is available in SettlementTracker to process invoice and generate invoice.

4.5.4.1. Contract Administration

4.5.4.1.1. Setup Meters

The meters at the connection point between the Arrass Solar PV Plant and NGSA transmission network will be setup in the SettlementTracker. This can be setup manually or uploaded to the systems using the standard meter import definition.
And solar irradiation data will be captured in the USD table. And table structure will be defined based on file format provided by Saudi IPP.

4.5.4.1.2. Setup Price curves definition for time series data

The following curves will be setup in the SettlementTracker to capture price and time-series data.

Name	Description	Source	Granularity	URL
SOFR	Secured Overnight Finance Rate	TBA	Daily	TBA
SAIBOR	Saudi Arabian Interbank Offered Rate for three-month SAR deposits	Reuters Thompson	Daily	Screen page “SUAA”
EXR	USD/SAR exchange rate	Saudi Arabian Monetary Agency (SAMA)	Daily	https://www.sama.gov.sa/en-US/FinExc/pages/currency.aspx
USPPI	USA Producer Price Index	U.S. Department of Labor Bureau of Labor and Statistics	Monthly	https://data.bls.gov/timeseries/PCUOMFG--OMFG--
KSACPI	Kingdom of Saudi Arabia Consumer Price Index	General Authority for Statistics Kingdom of Saudi Arabia	Monthly	https://www.stats.gov.sa/en/394

Table 11: Price Curves

4.5.4.1.3. Setup Counterparties

The following counterparties will be setup in the Online SettlementTracker.

Counterparty Name	Counterparty Descriptions	Counterparty ID	Counterparty Type
Arrass Renewal Energy Company	Arrass Renewal Energy Company	IPP	Internal
Saudi Power Procurement Company	Saudi Power Procurement Company	Offtaker	External

Table 12: PPA Online SettlementTracker Counterparties

The following counterparties will be setup in the Offline SettlementTracker.

Counterparty Name	Counterparty Descriptions	Counterparty ID	Counterparty Type
Arrass Renewal Energy Company	Arrass Renewal Energy Company	IPP	External
Saudi Power Procurement Company	Saudi Power Procurement Company	Offtaker	Internal

Table 13: PPA Offline SettlementTrackerCSS Counterparties

4.5.4.1.4. Setup Contract

Contract will be setup as non- standard in the SettlementTracker and required invoice line items will be mapped with formula for valuation.

Name	Description	ID	Currency	UOM	Status
Power purchase Agreement	Power purchase Agreement	PPA	SAR	kWh	Approved

Table 14: Contract

And following charges will be setup for invoicing under PPA Contract

PPA

- Total Payment Generation for Billing Period (TP_n)
- Energy Charge Generation (EC_n)
- VAT_n Due on TP_n
- Capital Cost Recovery component for Billing Period (CCR_n)
- Operations and Maintenance Cost Recovery component for Billing Period (OMR_n)
- Deemed Net Electrical Energy (NEE deemed) CASE 1
- Deemed Net Electrical Energy (NEE deemed) CASE 2
- Determination of Deemed Energy Coefficients
- Project Performance Ratio
- Actual Performance Ratio
- First Billing Period of Contract Year (logical rule)
- Determine number of Deemed Intervals from MDA interface report regarding Curtailment instruction and Deemed Net Electrical Energy Intervals notification.

The above charges and required formulae should be set up manually. Formula and charges may be changed which will be determined during implementation of Project.

Seq.	Title	Description	Acronym	Source Ref.	Formula	SettlementTracker Formula
1	Actual Net Electrical Energy	Actual net electrical energy for interval 'i' based upon MDA hourly metered quantity	$NEE_{i,act}$	Inferred	$NEE_{n\ act} = \sum_{i=1}^{I} NEE_{i,act}$	MeterVolm(1)+MeterVolm(2)+MeterVolm(X) Neei,act is equivalent to channel 'export' of meter
2	Cost Recovery Component	Cost Recovery component for Billing Period 'n' (month)	CCR_n	Sch 9 B.2	$CCR_n = CCR * Y_{ccr} * (EXR_n/EXR_r) + CCR * (1 - Y_{ccr})$	ContractRate(CCR)*ContractRate(Yccr)*(GetCurveValue(EXCHn)/GetCurveValue(EXRr))+ContractRate(CCR)*(1-ContractRate(Yccr))
3	O&M Cost Recovery Component	O&M Cost recovery component for Billing Period 'n' month	OMR_n	Scg 9 B.3	$OMR_n = OMR * Y_{omr} * (EXR_n/EXR_r) * (USPPI_n/USPPI_r)$	ContractRate(OMR)*ContractRate(Yomr)*(GetCurveValue(EXCHn)/GetCurveValue(EXRr))*(GetCurveValue(USPPIn)/GetCurveValue(USPPIr))+ContractRate(OMR)*(1-ContractRate(Yomr))*(GetCurveValue(KSACPIn)/GetCurveValue(KSACPIr))
4	Energy Charge	Energy Charge (cost for generation) for Billing Period 'n' (month)	EC_n	Sch 9.2.2	$EC_n = CCR_n + OMR_n$	ROW(2)+ROW (3)
5	Deemed Net Electrical Energy CASE1	Deemed NEt Electircal Eneeyg for interval 'I' (hour) based upon actual GPOA and applicable coefficients Case 1 before PCOD	$NEE_{i,deemed}$	Sch 17 Clause 2.4	$NEE_{i,deemed} = MAX(0, \left(\frac{IPNOM}{PNOM}\right) * (A_{1,n} * GPOA_i^2 + B_{1,n} * GPOA_i C_{1,n}) - NEE_{i,\ act}$	user defined function 2 deemed intervals defined by DI curtailment start and end times to flag intervals as 1 Curtailed or 0 not curtailed

Seq.	Title	Description	Acronym	Source Ref.	Formula	SettlementTracker Formula
6	Deemed Net Electrical Energy CASE2	Deemed Net electrical Energy for interval 'I' (hour) based upon actual GPOA and applicable coefficients CASE 2 after PCOD	$NEE_{i,deemed}$	Sch 17 Clause 2.4	$NEE_{i,deemed} = MAX(0, (A_{1,n} * GPOA_i^2 + B_{1,n} * GPOA_i + C_{1,n}) - NEE_{i,\ act}$	user defined function 1 deemed intervals defined by DI curtailment start and end times to flag intervals as 1 Curtailed or 0 not curtailed
7	Billing Period Deemed Net Electrical Energy	Deemed Net Electrical Energy for Billing Period 'n' based upon actual GPOA and applicable coefficients	$NEE_{n,\ deemed}$	Sch 17 Clause 2.4	$NEE_{n,\ deened} = \sum_{i=1}^{I} NEE_{i,deemed}$	Date effective ICOD < NOW = CASE 1 ELSE CASE 2 capture under counter party contract
8	Total Payment	Total Payment for Billing Period 'n' (month) for Plant export	TP_n	Sch 9 clause 2.1	$TP_{n\ gen} = ((NEE_{n,act} + NEE_{n,deemed}) * EC_n/100)$	[ROW(1)+ROW(7)]* ROW(4)/100
9	Value Added Taxation	Value added taxation amount determined by applying VAT rate (%) to Total Payment for Billing period 'n' (month)	VAT_n	Clause 9.8	$VAT_n = TP_n * (VAT_{RATE}/100)$	ROW(1)*ContractRate(VAT)/100
10	Grand Total Payment	Total Payment for billing period 'n' with addition of VAT charge for billing period 'n' (month)	GTP_n	Inferred	$GTP_n = TP_n + VAT_n$	ROW(8)+ROW (9) This functions sums VAT and TP regardless of whether TP is Export or Import
11	Total Consumption Payment	Total Consumption Payment for Billing Period 'n' (month) in respect of Plant metered import quantity of Net Electrical Energy	$TP_{n,con}$	Clause 6.5.2	$TP_{n,\ con} = NEE_{n,act,con} * ICT_n$	NEEn,act,con is equivalent to channel 'import' of meter

Seq.	Title	Description	Acronym	Source Ref.	Formula	SettlementTracker Formula
12	Actual Performance Ratio	Actual Performance Ratio		Sch 6 Clause 11.3	$APR = \dfrac{\frac{\sum_{i=1}^{j} Eact_i}{P_{mod}} * G_{stc}}{\sum_{i=1}^{i}(GPOA_i(1+\beta*(Tmod_m - Tmeas_i)))}$	
13	Projected Performance Ratio	Projected Performance Ratio		Inferred	$PPR = \frac{\sum_{i=1}^{j} Eact_i}{P_{mod}} * \frac{G_{stc}}{\sum_{i=1}^{i}(GPOA_i(1+\beta*(Tmod_m - Tmeas_i)))}$	
14	Coefficient A	Deemed Energy Coefficient B value for relevant month of relevant contract year	$A_{cy,m}$	Sch 17 clause 4.2	$c_n = GPOA_i$	SettlementTracker will make a function call to external program
15	Coefficient B	Deemed Energy Coefficient A value for relevant month of relevant contract year	$B_{cy,m}$	Sch 17 clause 4.2	$B_n = \frac{NEE_{i+1} - NEE_i}{GPOA_{i+1} - GPOA_i}$	SettlementTracker will make a function call to external program
16	Coefficient C	Deemed Energy Coefficient C value for relevant month of relevant contract year	$C_{cy,m}$	Sch 17 clause 4.2	$A_n = \dfrac{\frac{NEE_{i+2} - NEE_{i+1}}{GPOA_{i+2} - GPOA_{i+1}} = \frac{NEE_{i+1} - NEE_i}{GPOA_{i+1} - GPOA_i}}{GPOA_{i+2} - GPOA_i}$	SettlementTracker will make a function call to external program

Table 15: Formula Definit

Where:

I = Number of hours in the Billing Period

i = Metering Interval (number of the hour in day series)

j = the number of hours during the testing period

n = Billing Period (number of the month in year)

d = number of the day in the month series

D = Number of Calendar Days in Month

ODD = Number of days payment overdue

m = contract year

stc = standard temperature conditions

oft = Offtaker

mod = projected module temperature

meas = actual measured module temperature

act = actual metered generation value

deemed = value of generation or payment for generation based upon projection of generation from Solar irradiation measurements

r = reference value's period, date/time

prod = electricity producer

omr = operations and maintenance cost recovery

ccr = capital cost recovery

4.5.4.1.5. Global Solar Irradiation on the Plane of the Array

The Photovoltaic Panels deployed in the Arrass solar photovoltaic power plant are bifacial. This means that while the upper side is exposed to direct and diffuse solar irradiation as with a mono-facial panel the rear side will be exposed reflected irradiation.

The Global Solar Irradiation on the Plane of the Array for the settlement interval ($GPOA_i$) shall be determined as the average of the six Meteorological Station's pyranometers measures for In-Plane Irradiance plus Rear Side Irradiance.

$$AVERAGE\ (IN\text{-}PLANE_{MetStn\ 1\text{-}6,\ i}) + AVERAGE\ (REAR\ SIDE_{MetStn\ 1\text{-}6,\ i})$$

Equation 6: $GPOA_i$

Where:
$IN\text{-}PLANE_{Metstn\ 1\text{-}6}$ = PV SCADA data signals for the in plane of array pyranometer at meteorological stations 1 through 6 aligned with front of photovoltaic panels.
$REAR\ SIDE_{Metstn\ 1\text{-}6}$ = PV SCADA data signals for the in plane of array pyranometer at meteorological stations 1 through 6 aligned with rear of photovoltaic panels.

4.5.4.1.6. Define Invoice and Payment Date

The invoice date and payment date are configurable within the SettlementTracker so this can be defined during implementation.

IPP Invoice
The IPP invoice in respect of Net Electrical Energy produced by the Arrass Solar PV Plant.
Invoice Date: Within four (4) Business Days of the end of each Calendar Month (Billing Periods are calendar months).

SPPC Invoice
The SPPC invoice in respect of Net Electrical Energy consumed by Arrass Solar PV Plant.

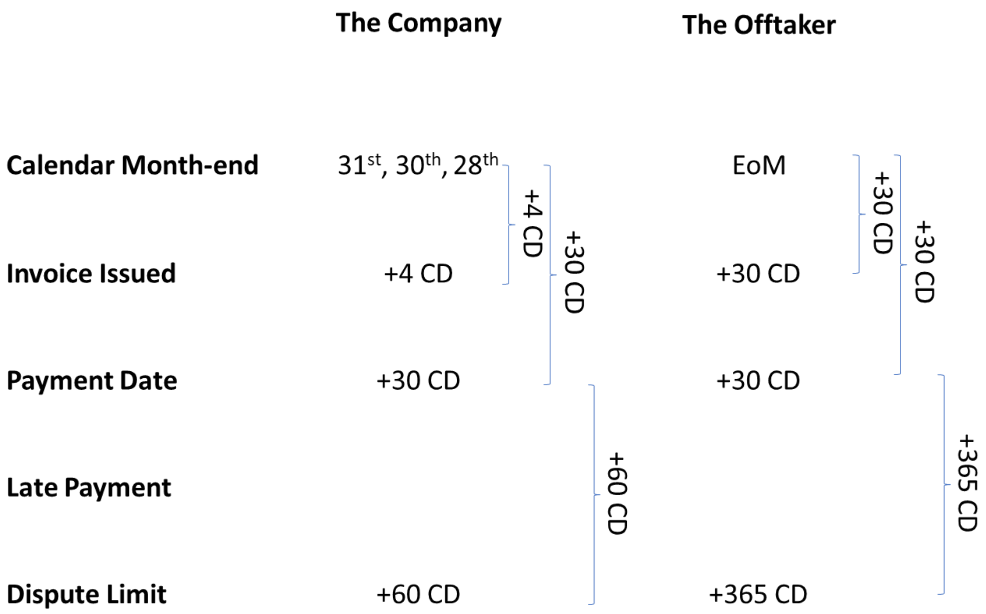


Figure 13: Invoice and Payment Calendar

Invoice Date: within thirty (30) days of the end of the Billing Period (calendar month) in which any such amounts accrue.

IPP and SPPC Invoice
Invoice Due Date: 30 Calendar Days from the date on which it is received by the counterparty.

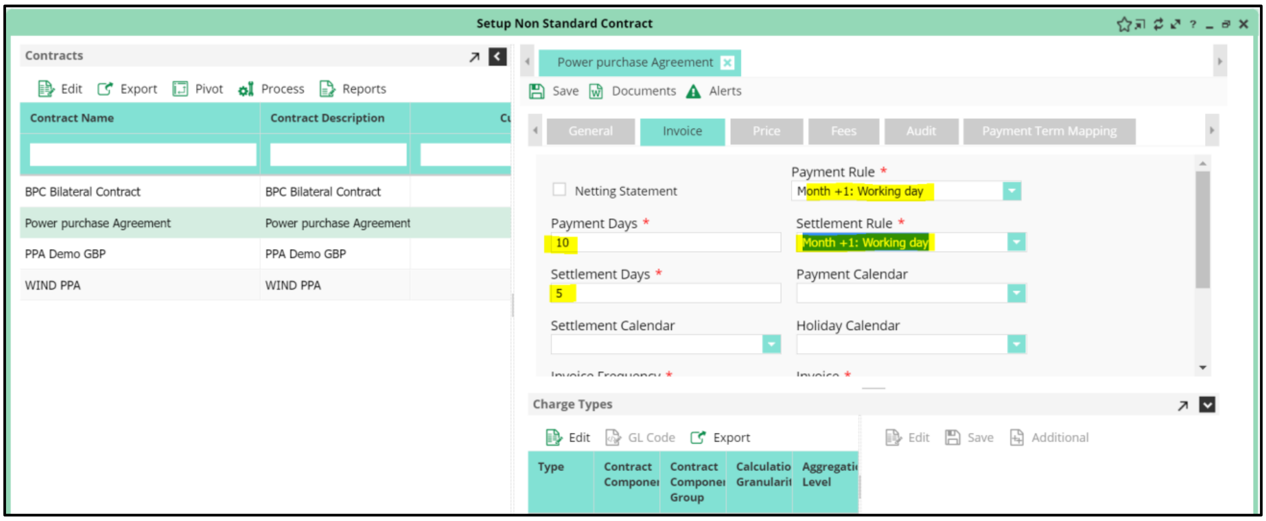


Figure 14: Setup Non-Standard Contract

4.5.4.1.7. Design Invoice Template

SettlementTracker has a standard PPA invoice template. And that standards will be deployed during implementation but if the client requires a custom ~~template,~~ then the Arrass team should provide a sample format before the start of the project and Hitachi team will design and implement during project at an agreed additional cost.


Sample Invoice format.pdf

Customer and Hitachi Energy to agree on a design for the Arrass invoice template during implementation sprints.

4.5.4.1.8. Compose Email

This is configurable in the SettlementTracker and can be configured during project implementation based on understanding between the Hitachi team and IPP.

Templates shall be agreed with the client and configured for the following suggested automatic email notifications.

- OID IPP to SPPC email
- VID SPPC to IPP email
- VID confirmation IPP to SPPC email
- EOD workflow IPP to SPPC email

4.5.4.2. Setup User Defined Tables

The Following user defined tables will be created in the SettlementTracker to captured different data required for invoice/ performance calculation. Additional tables might be added based on calculations during project implementation.

4.5.4.3. PPA Contract Term Projected NEE (Appendix A sch 17)

A User Defined table will be created in SettlementTracker to capture projected NEE.

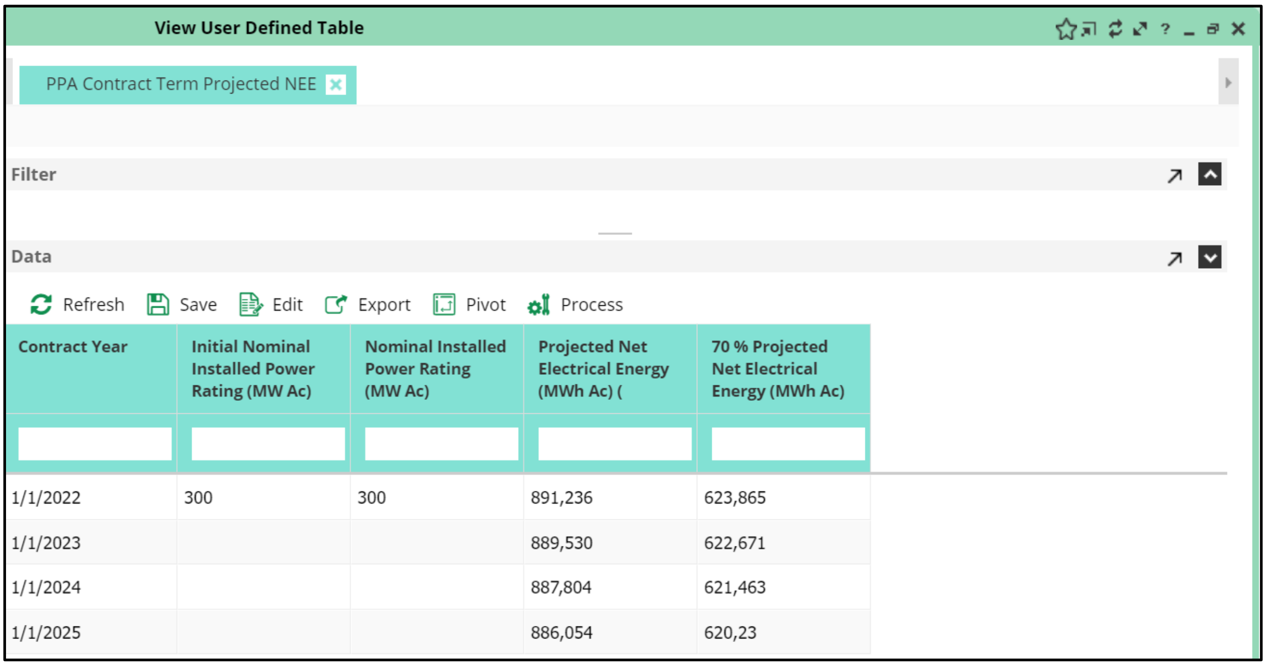


Figure 15: View User Defined Table

This can be setup manually or can be uploaded as bulk using import rule in the SettlementTracker.

4.5.4.4. Coefficients CY1 Table 1 in Schedule 17

User Defined Table will be created in SettlementTracker to capture Coefficients.

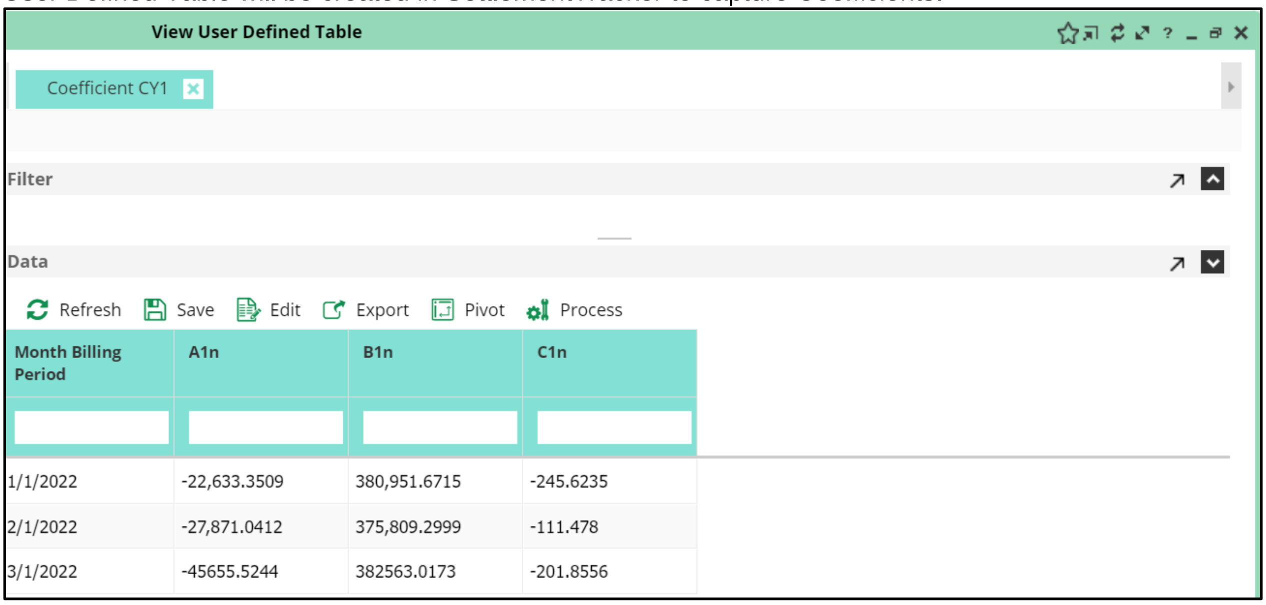


Figure 16: User Defined Table

This can be setup manually or can be uploaded as bulk using the import rule in the SettlementTracker.

- PPA Reference Data Schedule 9 reference data items

Short Name	Description
$KSACPI_{ref}$	Kingdome Saudi Arabia Consumer Price Index reference value
$USPPI_{ref}$	United States Producer Price Index reference value
EXR_{ref}	Exchange Rate USD: SAR reference value
CCR	Capital Cost Recovery component
Y_{ccr}	Foreign Proportion of capital cost recovery component

OMR	Operations and Maintenance cost recovery component
Y_{omr}	Foreign Proportion of O&M cost recovery component

Table 16: PPA semi-static data

- SPM static data

Name	Description	Value
P_{mod}	The sum of nameplate rating of the Arrass Solar PV plant's modules at Standard Temperature Condition (STC)	300 MW
β	The absolute (unsigned) value of the temperature coefficient from the PV module's data sheet for the module's nominal power output (in %/℃)	0.340%/℃[10]
$Tmod_i$	Projected monthly average module temperature (in ℃) as set forth in Form Sheet D of Schedule 1 of the PPA	Refer to Sheet D of Schedule 1 of the PPA
G_{STC}	Irradiance at Standard Temperature Conditions (STC)	1kW/m^2

Table 17: SPM Semi-static data

4.5.4.5. Guaranteed Performance Ratio

A User defined table (UDT) will be set up in SettlementTracker to capture Performance Ratio Guarantee (PRG) at both PCOD and ICOD. Although the PRG is the same at ICOD and PCOD for the Arrass project this may not be case for all projects, hence the requirement to capture these values separately in a UDT.

S.N	Month	Guaranteed Performance Ratio at Performance Testing (%)	Projected Monthly Average Module Temperature (Celsius)
1	January	91.9	37.5
2	February	88.3	42.4
3	March	83.2	45.6
4	April	80.7	48.4
5	May	80.6	51.1
6	June	81.9	50.8
7	July	81.3	50.5
8	August	84.2	50.2
9	September	84.6	51.1
10	October	87.3	48.2
11	November	89.5	43.9
12	December	90.0	42.8

Table 18: Performance Ratio Guarantee at ICOD.

S.N	Month	Guaranteed Performance Ratio at Performance Testing (%)	Projected Monthly Average Module Temperature (Celsius)
1	January	91.9	37.5
2	February	88.3	42.4
3	March	83.2	45.6
4	April	80.7	48.4
5	May	80.6	51.1
6	June	81.9	50.8
7	July	81.3	50.5
8	August	84.2	50.2
9	September	84.6	51.1
10	October	87.3	48.2
11	November	89.5	43.9
12	December	90.0	42.8

Table 19: Performance Ratio Guarantee at PCOD.

[10] Source: Temperature Ratings (STC) from Attachment 02 - LR5-72HBD-540M Data Sheet.pdf, amendment as notified by IPP

These performance ratio guarantees shall only apply during Acceptance Testing before PCOD. Therefore, the Actual Performance Ratio calculated during Acceptance Testing shall be compared to the performance guarantee during the relevant month when the Plant is being Acceptance Tested. These values are therefore for one-time use. In the event that the measurement period covers days in two different months then a weighted average (depending upon the number of measurement days in each month) of the respective Guaranteed Performance Ratio shall be used. For the avoidance of doubt, no additional adjustment shall be made to the measured Actual Performance Ratio in terms of measurement errors.

$$PRG_{wa} = \frac{a}{a+b} \times PRG_A + \frac{b}{a+b} \times PRG_B$$

Equation 7: Bi-month Performance Ratio Guarantee

Where:
a = number of measurement days in month A
b = number of measurement days in month B
PRG_A = Performance Ratio Guarantee for Month A
PRG_B = Performance Ratio Guarantee for Month B
PRG_{wa} = performance Ratio Guarantee weighted average for measurement period

In subsequent Contract Years the degradation of the module performance shall mean that the actual performance ratio can be expected to progressively decline year on year from these values.

4.5.4.6. Weather Forecast Data

A standard table in SettlementTracker is available to capture the day-ahead and within day forecast weather data required for use in determining the projected generation for day-ahead and within day generation forecasts.

4.5.4.7. Availability-Energy Notice

A user defined table (UDT) shall be created in SettlementTracker to capture the day-ahead and any superseding availability-energy notices.

Station ID	Notice Date	Start Date	End Date	Duration	Reason	Availability
VarChar	DateTime	DateTime	DateTime	Float	Varchar	Float

Table 20: Availability-Energy Notice

4.5.5. Interface-Inbound/Outbound

4.5.5.1. Inbound

4.5.5.1.1. Meter Data

Meter Data will be uploaded via ODBC in SettlementTracker, and, in case of manual upload, standard meter import format should be used and upload it using standard meter data import rule.

4.5.5.1.2. Weather Data

Weather Data will be uploaded via ODBC in SettlementTracker, and in case of manual upload, the standard weather data import format should be used and upload it using the standard weather data import rule.

4.5.5.1.3. User Defined Table

Required data will be uploaded in SettlementTracker via flat file or from folder. This process can be defined during project implementation.

4.5.5.1.4. Price and Time series Data

Required price and time series data will be uploaded in SettlementTracker. For manual import, start price curve format should be used and upload it using standard price curve data import rule.

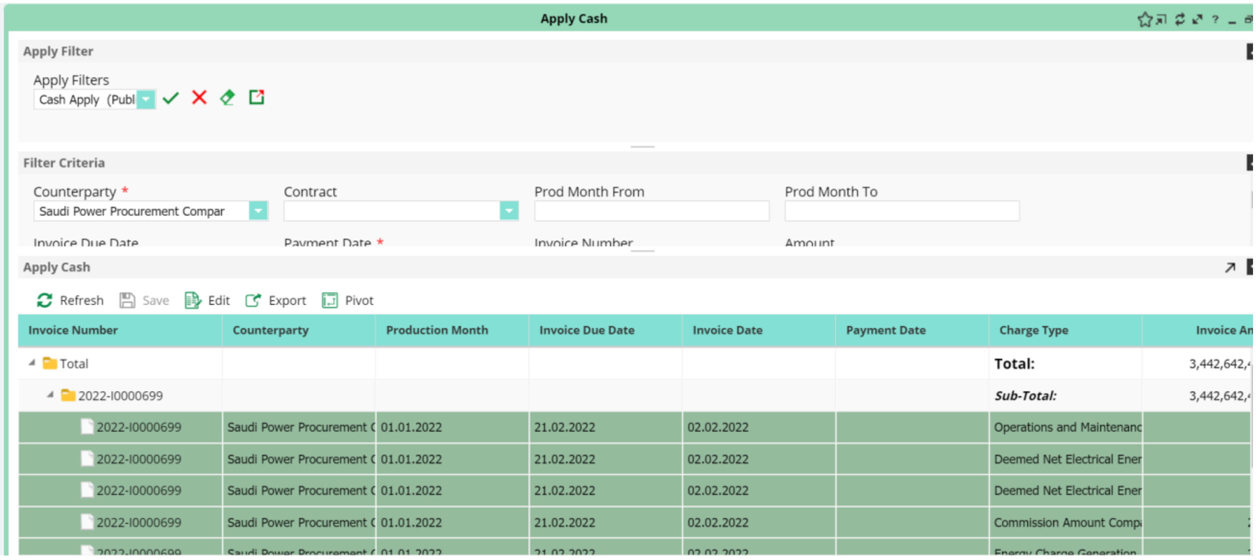
4.5.5.1.5. Availability-Energy Notice Data

Availability-Energy notice data shall be imported via manual upload, a custom availability import format should be used together with a custom availability data import rule.

Availability-Energy
%20Notice%20ISPEC

4.5.5.1.6. Cash Received Data

Cash received information shall be manually entered through the application's using 'Apply Cash" GUI in the SettlementTracker.



4.5.5.2. Outbound

List of outbound

S.N.	Report Title	Report Description	Category
1	SPPC Invoice	Invoice from SPPC to IPP for monthly electricity exported from Solar Plant	Custom
2	IPP Invoice	Invoice by IPP for monthly electricity Generated by Solar Plant	Custom
3	Backing Sheet SPPC Invoice	Supporting hourly detail for monthly SPPC invoice	Custom
4	Backing Sheet IPP Invoice	Supporting hourly detail for monthly IPP invoice.	Custom
5	Actual Performance Ratio (APR)	APR compared to Guaranteed Performance Ratio from Section D.2 of Form Sheet D.	Custom
6	Projected Performance Ratio	PPR compared to Guaranteed Performance Ratio from Section D.2 of Form Sheet D.	Custom
7	Comparison of Projected NEE vs Actual NEE	Report on Actual and Projected Net Electrical Energy values.	Custom
8	Coefficients A, B and C	Report of monthly coefficient A, B and C's values for contract year.	Custom
9	Forecast values for projected NEE (day ahead)	Day-ahead generation forecasts	Custom

S.N.	Report Title	Report Description	Category
10	Forecast values for projected NEE (intraday)	Intraday generation forecasts	Custom
11.	Forecast values for Deemed NEE (24 hours)	Deemed generation forecasts	Custom
12	Holiday Calendar Report		Standard
13	Trader Definition Report		Standard
14	Price Curve Definition Report		Standard
19	Contract Definition Report		Standard
20	Location Definition Report		Standard
21	Counterparty Definition Report		Standard
22	Counterparty Contract Definition Report		Standard
23	Counterparty Contact Definition Report		Standard
24	Forecast Profile Definition Report		Standard
25	Meter Definition Report		Standard
26	Commodity Definition Report		Standard
27	Report Privilege Report		Standard
28	User Extract Report		Standard
29	Power Forecast Data Ahead		Custom
30	Original Input Data Report	Comprising determinants necessary to calculate settlement invoices ref. 4.3.2	Custom
31	Validated Input Data Report	Comprising validated determinants necessary to calculate settlement invoices ref. 4.3.2	Custom
32	Meter Data Report	MDA Metered Data Quantities report for both Imported and Exported electricity at the connection point	Custom
33	Meteorological Data Report	Report of meteorological data values captured by PV SCADA system e.g., ambient temperature, module temperature, dew point, wind speed, humidity, cloud cover, wind direction, wet bulb	Custom
34	SCADA net electrical energy	Report on Net Electrical Energy and Power Factor values captured by PV SCADA.	Custom
35	Irradiation Report	Report of Solar irradiation values captured by PV SCADA for DHI, DI, DI and GPOA	Custom
36	Sunrise sunset time report	Report on predicted Sunrise and Sunset times for Arrass Solar PV Plant.	Custom
37	Comparison OID and VID report	Comparison of difference between OID data values and VID data values	Custom
38	Shadow SPPC Settlement Report	Shadow Invoice values calculated by SPPC of the charges invoiced by IPP.	Custom
39	Shadow IPP settlement Report	Shadow Invoice values calculated by IPP of the charges invoiced by SPPC.	Custom
40	Availability Report	Monthly report for hourly comparison of day ahead availability with actual metered generation.	Custom
41	Cash Report	Monthly report of cash postings	Standard

Table 21: Standard and Custom Reports

Standard reports will be enabled and those can be used to view data available in the systems or outbound export as well.

4.5.6. Process Invoice/Generate invoice

Invoice should be manually processed in the SettlementTracker end of every Month. This section discusses requirements and implementation approach for processing invoices and send approval for manager. Process and view invoice UI will be used available in SettlementTracker to process invoice and generate invoice.

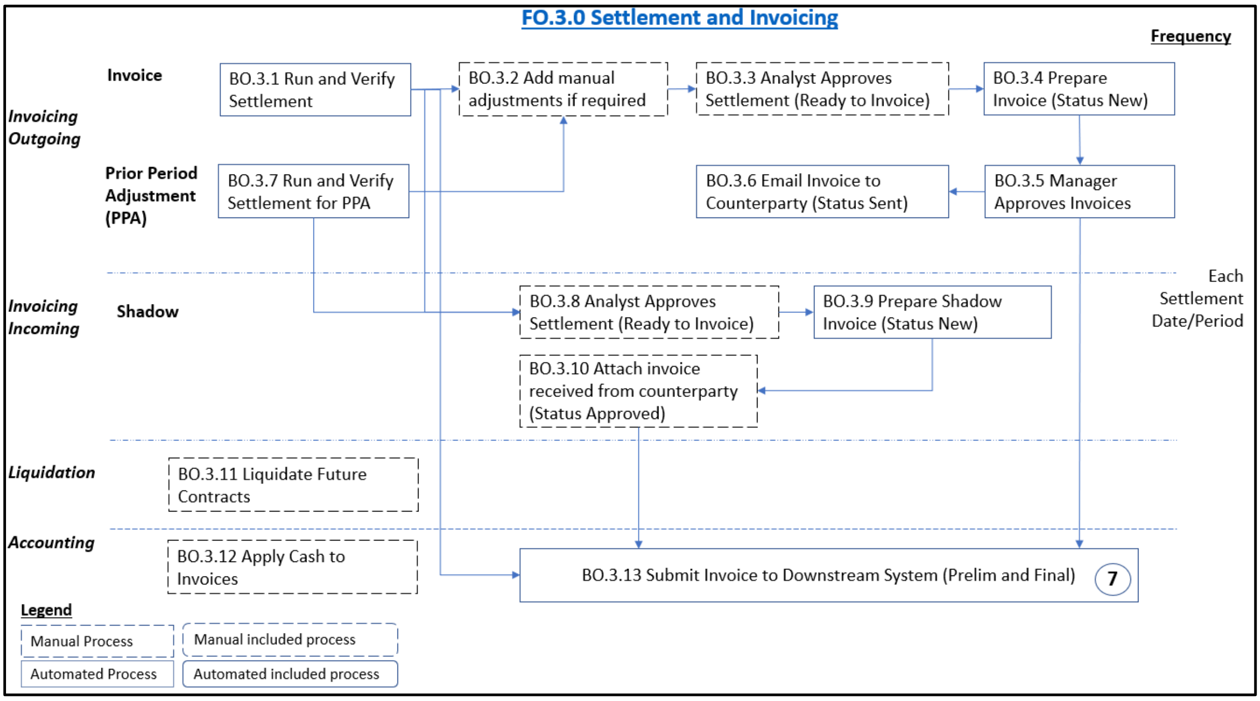


Figure 17: Front Office Settlement and Invoicing Process

4.5.6.1. Invoicing Process

- Required input should be verify manually before process an invoice (BO.3.1)
- Invoice Can be manually adjust for the minor variance before sending to counterparty and, also calculate prior month variance and include in the current month invoice (BO. 3.2 and (BO. 3.7)
- Contract Analyst and Contract manager approved invoice and sent it to counterparty (BO. 3.3, BO. 3.4, BBO.3. 5, BO. 3.6)
- Invoices will be emailed to the counterparty automatically (BO.3.6). Invoices can be created in PDF format.

4.5.6.2. Add Manual Adjustment If required.

- If required, manual adjustments will be added. Manual adjustments are those charges the system may not calculate. Manual adjustments can be entered to make rounding corrections, add taxes or other line items.

4.5.6.3. Prior period Adjustment

- True up feature will be enabled in the SettlementTracker and should be used to calculate prior month variance.

4.5.7. Workflow Alerts and Notifications

Standard invoice workflow will be enabled. Contract Analyst manually run and verify invoice amount and change status to "Ready for Invoice" and below workflow will be trigger automatically.

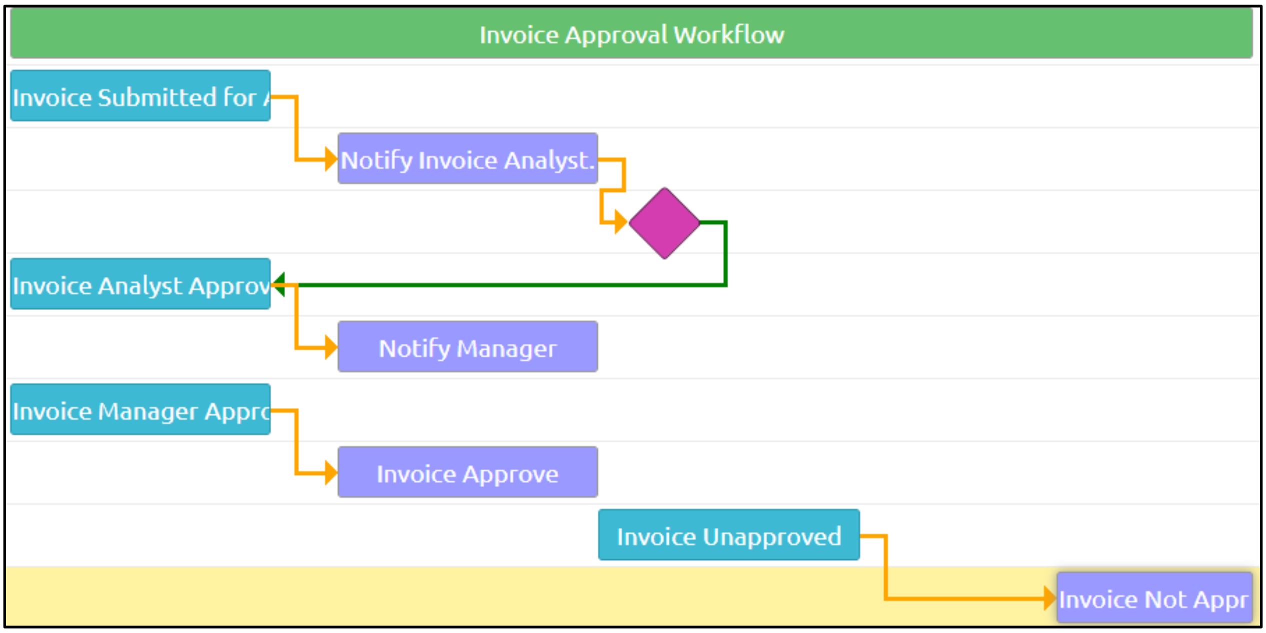


Figure 18: Standard Invoice Workflow

4.5.7.1. Accounting

- Apply Cash to Invoices (BO.3.12)

5. Enhancements Assessment and Gap Closure Plan

The following enhancements/gaps will need to be resolved:

Area	Enhancement	Description / Approach
Generation Forecast	Forecast Tool	SettlementTracker Module trained to forecast electricity production implemented within SettlementTracker
Coefficient determination	External Coefficent Calculator	Function call to external dedicated coefficient calculation program

Table 22: Enhancements Assessment and Gap Register

APPENDIX A: ASSUMPTIONS

#	Assumption Title	Assumption
1	Generation Forecast	The generation forecast shall be the Day-Ahead forecast of NEE production by the Arrass Solar PV Plant as measured at the Connection Point with hourly granularity for the Day immediately following the current date. SPM shall produce forecasts for all non-business days until the next Business Day+1 so that on return-to-work next forecast produced is for the Day Ahead), and for the Intraday forecast of NEE production by the Arrass Solar PV plant being for a period of three (3) hours ahead during the current dispatch day, with a granularity of 1 hour. This forecast shall be repeated every hour with the 3-hour period rolling forward 1 hour each time, or as weather forecast availability permits.
2	Generation Forecast Day-ahead	The generation forecast Day-Ahead (DA) report file shall be produced by the PSS. The client is responsible for exporting the data and submitting it to the Grid Operator.
3	Generation Forecast Intraday	The generation forecast Intraday (ID) report file shall be produced by the PSS. The client is responsible for exporting the data and submitting it to the Grid Operator.
5	Original Input Data	The original input data (OID) for validation by SettlementTracker shall include only that data essential to calculate the settlement invoices. The validation of plant operational and planning data shall be managed by the clients' other systems.
6	SettlementTracker TimeSeries Daily data Update	The SettlementTracker systems shall import Daily the Exchange Rate and Interest Rate data to the TimeSeries data store within SettlementTracker
7	SettlementTracker TimeSeries Monthly data Update	The SettlementTracker systems shall import Monthly the Inflation indices data to the TimeSeries data store within SettlementTracker
8	Force Majeure declaration	The SettlementTracker systems shall capture the start and end dates of a counterparty's declaration of either Force Majeure or an Offtakers Risk Event in accordance with their PPA.
9	Internal/External Counterparty configuration in Online SettlementTracker and Offline SettlementTracker.	The IPP (Company) Counterparty shall be configured as the Internal Counterparty in the Online SettlementTracker While the SPPC (Offtaker) shall be configured as the External Counterparty in the Online SettlementTracker
10	Backing Sheets	Detailed backing sheets will be required for each monthly invoice providing hourly granularity data for the whole billing period and settlement charge determinant components with an indication of the calculations used to derive resulting charges
11	Industrial Consumption Tariff	The industrial consumption tariff rate shall be a single rate tariff as specified at the following URL Http://se/com/sa/en-us/Customers/Pages/TariffRates.aspx
13	Outage Settlement	Scheduled and forced outages shall be settled by PSS in accordance with the specific rules detailed in the PPA and Schedules 9 and 17, such that during curtailment periods the deemed energy payment shall include generation capacity which is declared to be in outage.

#	Assumption Title	Assumption
14	Deemed Delivery Rules	That there is a Deemed Delivery rule in KSA law such that an invoice emailed to a counterparty before the Government Office close of Business hour shall be deemed delivered on the same day. The business process of the invoicing party shall ensure that invoices are finalised, generated and emailed to the counterparty before the Government Office close of business hour. This will enable the calculation of payment due date at the time of Invoice Generation and thus the determination of interest (commission) charges on early or late payments of owed amounts.
15	Cash Received Amount	The amount of cash received in respect of the previous billing period shall be manually entered through the SettlementTracker GUI.
16	KSACPI values	That the KSACPI values to be used as reference data shall be that of the “General Index” sourced from the General Authority for Statistics Kingdom of Saudi Arabia’s website URL https://www.stats.gov.sa/en/394

Table 23: Assumptions

APPENDIX B: Glossary

#	Title	Acronym	Description
1	Actual Performance Ratio	APR	The outturn Ratio of actual and theoretically possible energy outputs
2	Azure Site Recovery	ASR	A disaster recovery as a service (DRaaS) offered by Azure for use in cloud and hybrid architecture.
3	Backup Metering System	BUMS	A secondary metering system in case of a fault in the primary metering system
4	Billing Period		The Billing Period under the PPA shall be 1 hour (60 minutes)
5	Capital Cost Recovery	CCR	Capital Cost Recovery component of energy charge
6	Central Settlement System	CSS	The PPA settlement system of the Offtakers
7	Chart of Accounts	COA	A chart of accounts (COA) is a financial organizational tool that provides a complete listing of every account in the general ledger of a company, broken down into subcategories.
9	Dispatch Load Center	DLC	The operational control center for the transmission system of the National Grid Saudi Arabia
10	End of Day	EOD	End of business day typically used to indicate a business process which occurs at the end of each business day.
11	Energy Charge	EC	The charge for electricity generated under the PPA
12	Exchange Rate	EXR	The value of one currency for the purpose of conversion to another. In this instance USD to SAR
13	Factory Acceptance Test	FAT	System testing of software application before UAT.
14	Global Irradiation on the Plane of the Array	GPOA	The irradiation in kWh/m^2 measured per each metering interval 'i' with an on-site pyranometer with an identical inclination as the PV modules
15	Independent Power Producer	IPP	An IPP is an entity that is not a public utility but owns facilities to generate electric power for sale to utilities and end users.
16	Initial Commercial Operation Date	ICOD	
17	Interim Nominal Installed Power Rating	IPNOM	The nameplate rating of all currently installed inverter AC power at 50 °C
18	Kingdom of Saudi Arabia	KSA	A country on the Arabian Peninsula in Middle East.
19	Kingdom of Saudi Arabia Consumer Price Index	KSACPI	An index of the variation in prices for retail goods in the Kingdom of Saudi Arabia
20	Metering and Data Acquisition System	MDA	A system that collects settlement quality meter data on the generation and consumption of electricity between suppliers and customers for use in contractual settlement.
21	Net Electrical Energy	NEE	The Net Electrical Energy metered at the connection point of Arrass Solar PV Plant to the NGSA transmission grid (measured in kWh over each metering interval (hour). Note this value may be either positive or negative as NEE value may be imports or exports across the connection. Although normally the calculation would be scheduled for a period when the plant's output has not been curtailed.
22	Neural Network		A series of algorithms that endeavour to recognise underlying relationships in a set of data through a process that mimics the way the human brain operates.

#	Title	Acronym	Description
23	Nominal Installed Power Rating	PNOM	The nameplate rating of all installed inverter AC power at 50 °C
24	Object Linking and Embedding	OLE	A Microsoft technology that facilitates the sharing of application data and objects written in different formats from multiple sources. Linking establishes a connection between two objects and embedding facilitates application data insertion.
25	Offtaker		The PPA counterpart acting as the purchaser of the energy.
26	operational Maintenance Cost Recovery	OMR	Operations and Maintenance Cost Recovery component of energy charge
27	Original Input Data	OID	The determinants necessary to calculate the settlement charges for the PPA.
28	Output Metering System	OMS	The primary electronic electricity metering system at the Connection Point
29	Photovoltaic	PV	the generation of voltage when radiant energy falls on the boundary between dissimilar substances (such as two different semiconductors)
30	Photovoltaic Supervisory Control and Data Acquisition	PV SCADA	A control system architecture comprising computers, networked data communications, RTU (Remote Terminal Units), PLC (programable Logic Controllers), PPC (Power Plant Controller), sensors and a GUI (Graphical User Interface) for high level supervision of a Photovoltaic Power Station
31	Plane of Array	POA	indicates solar irradiation measurement taken in the same orientation as the PV panels. The orientation is described by vectors of the array tilt angle and the array azimuth angle.
32	Planned Commercial Operation Date	PCOD	The Planned date for the commencement of commercial operation of the Arrass Renewable Energy Company's solar power plant
33	Plant Accounting and Settlement System	PSS	The module of PSS is responsible for determining the contract settlement values and invoices. Also referred to as the TCM.
34	Plant Performance Monitor	PPM	Plant Performance Monitor also referred to as the SPMB55B53:D61B50:D61B48:D61B46:D6B2:D61
35	Plant Settlement System	PSS	Both the ESS and CSS software systems combined.
36	Power Purchase Agreement	PPA	A long-term electricity supply agreement between two parties, usually a power producer and an Offtaker
37	Price Curve		The time series price data for a specific product e.g., electricity delivered to the NGSA or the USD: SAR Exchange Rate over time.
38	Saudi Arabian Interbank Offer Rate	SAIBOR	Saudi Arabian Interbank Offer Rate for three-month SAR deposits
39	Saudi Arabian Riyal	SAR	The official currency of the Kingdom of Saudi Arabia
40	Saudi Power Procurement Company	SPPC	The Offtaker
41	Arrass Renewable Energy Company		The IPP
42	Standard Test Conditions	STC	Industry standard for the conditions under which solar panels are tested.
43	Structured Query Language	SQL	An acronym for Structured Query Language, which lets users access and manipulate databases
44	System Performance Monitor	SPM	The module of PSS responsible for determining the APR. Also referred to as the PPM.

#	Title	Acronym	Description
45	Tariff Calculation Mechanism	TCM	Tariff Calculation Mechanism also referred to as the PSS
46	Total Payment	TP	The total charge for electricity for a billing period
47	Unit of Measure	UOM	The standard unit of measurement
48	United States Dollar	USD	The official currency of the United States of America
49	United States of America Producer Price Index	USPPI	A monthly measure of change in the prices received by domestic producers in the USA
50	User Acceptance Test	UAT	The final stage of any software development lifecycle before go-live. Its purpose is to determine that the software does what it is designed to do in real-world situations. Actual users test the software to determine if it does what it was designed to do.
51	Validated Input Data	VID	The reviewed and amended OID
52	Value Added Tax	VAT	Value Added Tax is a flat tax levied on an item

Table 24: Glossary

6. Project Experience

SettlementTracker is used by our customers as part of integrated energy trading and risk management systems (ETRM) as well as on a standalone basis where it would integrate with 3rd party systems. Below is a sample of customers that are using SettlementTracker as a specialised module:

Customer	Objective	Why	Results
essent	Higher flexibility to handle unique and evolving needs. Lower Total Cost of Ownership (TCO). The rapid delivery process allows for the quick replacement of existing systems.	**SettlementTracker** was best able to model the company's most complex scenarios. SettlementTracker's flexible architecture enabled Hitachi Energy to commit to quickly building out the necessary capabilities to bridge the few identified project gaps. Hitachi Energy product team had demonstrated a superior understanding of Essent's business problems and requirements.	Essent could quickly translate their business needs into the system configuration and develop additional functionality without a protracted development process. Improved system flexibility and increased efficiency and effectiveness.
Gasum	Due to market unbundling, Gasum must prepare to scale up NG, LNG, and Power trading. A flexible, scalable, and easy to integrate into the new IT landscape. Regulatory reporting functionality for REMIT, EMIR and MiFID2.	TRMTracker.com offers a comprehensive **cloud-based** C/ETRM Software-as-a-Service solution that is flexible, scalable and easy to integrate into the new IT landscape, while seamlessly expandable to enterprise-level functionality down the road.	The system was configured with based implementation completed by September 2019.
ACWA POWER أكوا باور	Implement a comprehensive system to support ACWA Power Turkey's existing & planned commercial strategy Enable flexibility to adapt to changing	Hitachi gave us the confidence with their technology and implementation approach to achieve these important business objectives.	An ETRM supporting the trading and asset optimization for the Turkish generation capacity, in the DA, Intraday, and BM incl. an interface with

Customer	Objective	Why	Results
	market conditions and new strategies.	ACWA Power Turkey will benefit from TRMTracker's enterprise-level functionality that Hitachi Energy's flagship C/ETRM system provides	Exchanges (EPIAS) and support for BGM
Eneco	Improve efficiency and effectiveness by automating the back-office processes. Settle and account for most of its energy contracts (standard and custom) in a single system. Reduce back-office cost, improve financial accuracy down to the cent, and thereby maximise operational margin.	**SettlementTracker** provides unparalleled configuration ability of contract rules and billing determinants that is template-based, and formula-driven. Proven integration with other ETRM systems Hitachi engages like a partner.	>30% Reduction in effort for invoice transactions. 100% transparency on charging data to applicable large commercial clients.
NAFTOGAZ	Establish a single database of counterparties and physical gas trading agreements. Reengineering and automation of physical gas trading business processes. Automate procedures for controlling risk limits and powers. Automate modern tools for assessing and modelling commercial risks. Integrate ETRM system with internal systems of Naftogaz Group and external platforms to enable users to operate in a	TRMTracker is a state-of-the-art ETRM system, a "live" real-time and highly flexible ETRM system, delivering intuitive workflow, innovative reporting, and industry-leading integration capabilities.	Phased approach: Front Office Live. The contract approval period was reduced to 15 minutes.

Customer	Objective	Why	Results
	single environment and reduce.		
enercity	A future-proof system and integration of new applications into the already existing application landscape. Establishing a highly automated trade life cycle, to enable traders to focus mainly on trading not on “post-deal” management, including high automation of deal capture, back office, and risk control processes. Optionally, the replacement of the existing scheduling and nomination application. Introducing workflow processing including automated process monitoring.	TRMTracker system can deliver highly automated front-to-end trade life cycle management, risk management, and position management, that could easily be integrated into the existing application landscape.	Digitalization and automation of the front to back energy trading **and settlement process**
Xcel Energy	Administering the wholesale electric sales contracts, settlement, billing, and accounting for Xcel Energy’s three operating companies: Northern States Power Company Minnesota, Northern States Power Company Wisconsin and the Public Service Company of Colorado	**SettlementTracker** was uniquely qualified to support wholesale electric contracts that are unique and complex and are modelled using user-configurable formulas.	A straight-through processing platform was implemented to automate meter data and billing determinants, settlement, and automatic accounting entries.

Customer	Objective	Why	Results
RWE	Automation of hedge accounting effectiveness calculation, fair-value and MTM accounting. Effectiveness testing support Dollar Offset and Regression testing. Lower Total Cost of Ownership (TCO). The rapid delivery process allows for quick replacement of the existing system.	Available as a stand-alone system that seamlessly can interface to existing systems. FASTracker had the capability and technical infrastructure that could be customised to address RWE Supply & Trading's unique requirements.	More than 20% improvement in business process efficiency. An integrated platform Standard solution across Supply & Trading organisations in different geographies.

Figure 4: Selected Client Experience

7. System Description

This section provides details of the offered software (SettlementTracker) covering features, architecture, hardware architecture, access and security and interface.

7.1. About SettlementTracker

SettlementTracker is a stand-alone settlement and accounting module specifically designed to automate and manage complex energy contracts, settlements, and billing. It speeds the time to invoice while minimising operational and dynamic workflow controls. The system automates all processes from meter data and billing determinants management to settlement and shadow settlement with counterparties. From transaction capture to invoicing, the system will improve the time to invoice and minimise invoice and accounting errors.

SettlementTracker supports the straight-through-processing (STP) of complex bi-lateral and shadow settlement processes. In addition, the next-generation architecture supports ease-of-integration with Energy Trading systems, accounting systems and external data feeds such as Meter Data Management Systems.

Featuring business intelligence reporting, automatic alerts, a dynamic user message board, compliance management, and full audit trails, SettlementTracker's workflow automation allows for seamless processing of transaction lifecycles across multiple departments.

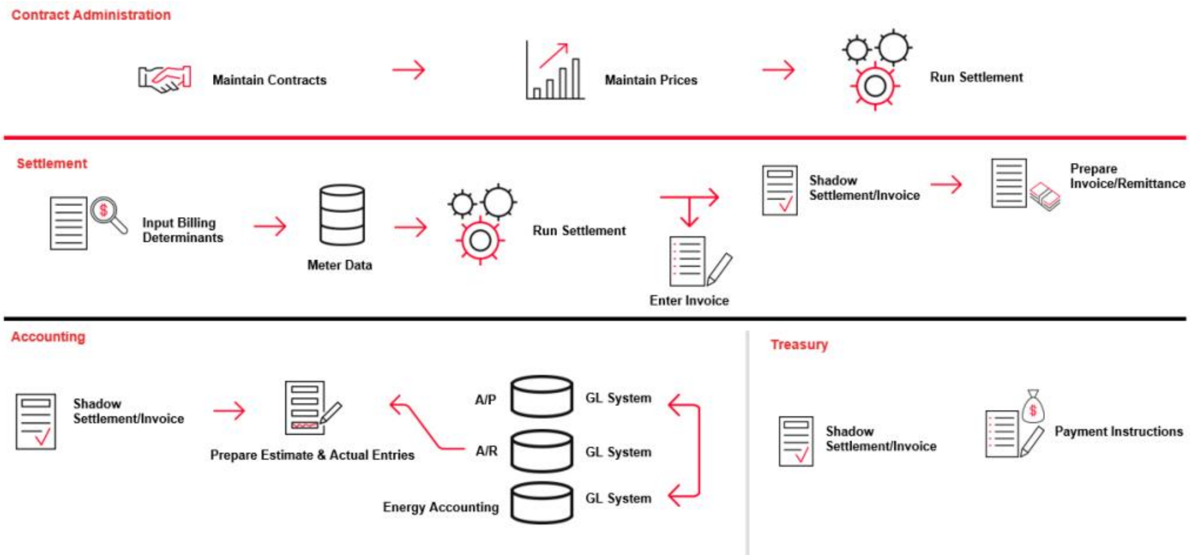


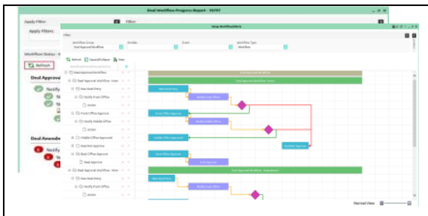
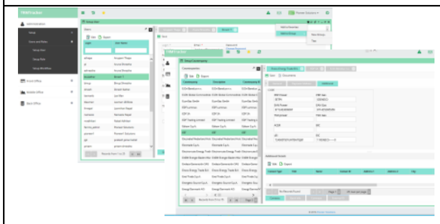
Figure 5: Overview of SettlementTracker

- Minimise financial risks by automating settlement and accounting processes
- Contracts Administration
    - Capture all contracts and charge types
    - Support for user-configurable business logic
    - Support for price fixations and un-fixations
    - Support for cross-commodity and time-based pricing
    - Support for on-peak/off-peak hour blocks
    - Define/manage billing determinants
- Settlement
    - Import meter and pricing data from external sources

- Settlement and shadow settlement support
  - Estimate and finalise deal and contract settlement
  - Cash flows and P&L reporting
  - Standard reports, ad-hoc report writer and OLAP cube support
  - Manual and automated prior period adjustments
- Invoice & Accounting
  - Capture/shadow incoming invoices and variance analysis and dispute resolution
  - Easily prepare invoices, remittance statements and detailed supporting documents
  - Send payment instructions electronically
  - Fully integrated settlement processing
  - Prepare estimates and easily finalise entries
  - Auto upload accounting entries into GL system(s)
  - Accrual Journal Entry Report
  - Reconcile Cash Entries for derivatives

SettlementTracker is web-based and template and formula-driven to offer a next-generation platform that is both easy to configure and easy to use. The scalable thin-client architecture is simple to set up and maintain and allows for the system to easily grow and evolve with your business needs for years to come - resulting in a lower overall cost of ownership and better efficiency and consistency for your business.

7.2. Innovative Features

	• The system provides an extensive workflow setup for approval and notifications. • Easy set up with intuitive workflow diagram. • Calendar function can be used with integration to MS-Outlook
	The system has a high degree of user-configuration ability (e.g., templates, formulas, book structures, and reporting). In case the right functionality cannot be achieved this way, Hitachi Energy can customise the system. The following are examples, but not limited to, of extensions that are possible by users. When this approach is used, a new upgrade will have no impact on the Customer. • User-Defined Fields and User Defined Tables can be added to extend data models such as on Deals, Counterparty, Contracts etc. • User-defined formula logic can virtually model any settlement business rules. • The mapping table allows for users to easily add new tariffs, levy or taxes that then can be referenced in the formula • Time series data will allow the user to add the required number of new time series data tables and easily import data

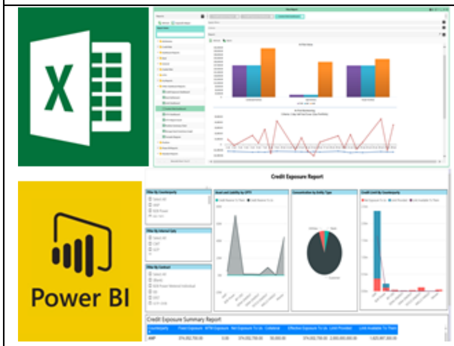
	• Import/Export rule allows users to easily import data from external sources/files by mapping fields and defining other rules • The report manager will allow the user to create new reports or export data. Excel add-on capability will allow users to build additional reports if this approach is approved by the Customer, IT organization.
	• In addition to standard Reports and an embedded Reportwriter, the system provides an Excel Add-In and integration with Microsoft Power BI • All system reports are available via Excel Add-In or Power Bi via a direct connection to the database, so data used for analysis is always up to date • Reports created in Excel Add-in or Power Bi can be easily deployed into the CTRM system (as if it is a native report in the CTRM system) and shared with other users.

Figure 6: Innovative features

7.3. Architecture Overview

Our software applications, including the proposed system solutions, run on our FARRMS (Financial and Regulatory Risk Management System) platform. FARRMS employs scalable multi-tier web-based client/server architecture based on Microsoft technology and offers several industry-standard solutions.

Our solution is easy to integrate with any external system/process. Given it is based on a thin, web-based client-server architecture, users only require a web browser to access the system regardless of when deployed on-premises or when Hitachi Energy hosts this remotely.

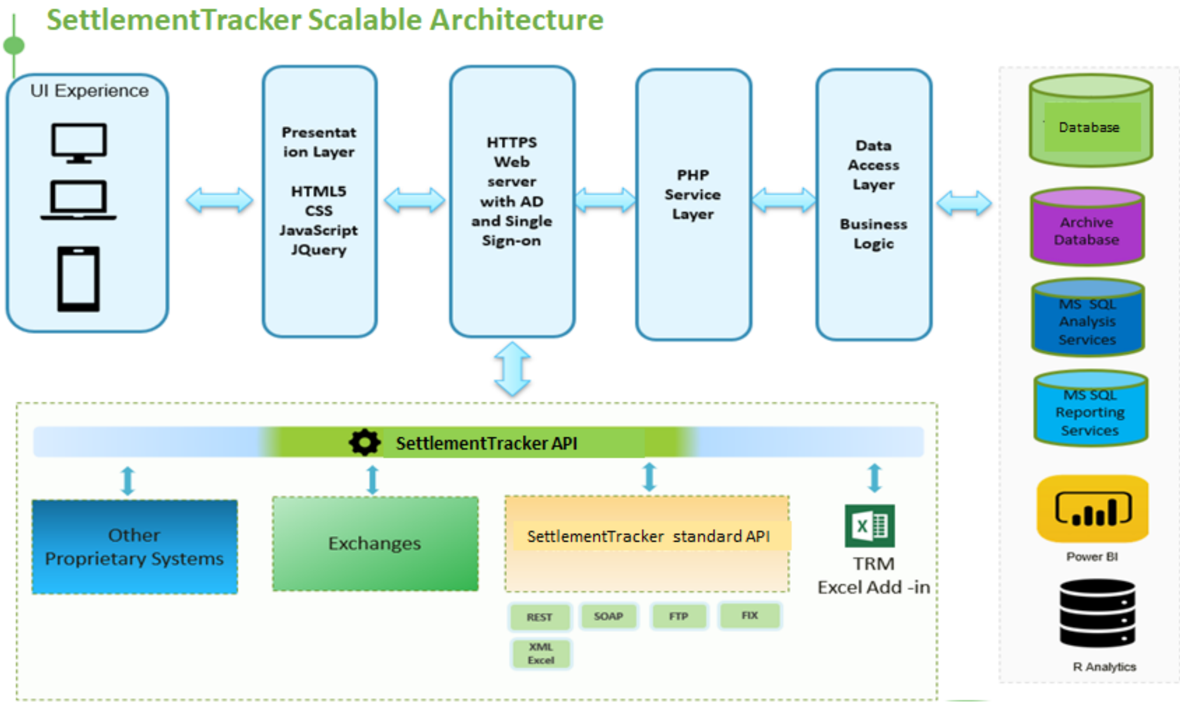


Figure 7: Architecture overview

7.3.1. Hardware Infrastructure

Recommended server requirements are presented in the table below.

Description	Web Server	Database Server	Proxy Server
CPU	Quad core 2.30 GHz	Hex core 2.30 GHz	Dual core 2.30 GHz
Memory	16GB	64GB	4GB
Storage disk	50 GB	500 GB	25 GB
Microsoft Windows Server	Windows Server 2012	Windows Server 2012	Linux
Microsoft SQL database	NA	SQL Server 2016 or Higher – Standard Edition*	NA
PHP	PHP 7.3.9	NA	NA
IIS	IIS 7 or later	NA	NA
Microsoft .Net Framework	.Net Framework 4.0	.Net Framework 4.0	NA
Microsoft Excel	NA	Office 2010 or higher	NA
*For high availability environment, SQL Server enterprise edition is required.			

Figure 8: Overview of the server environment and disaster recovery

Recommended client requirements are presented in the below table.

Descriptions	Specifications
Laptop/Desktop	Windows 7 or higher
Browser	IE 10 or higher, Chrome
Screen Resolution	1024 x 740 or higher
Network Bandwidth	100 MBPS
Microsoft .Net Framework	.Net Framework 4.0
Microsoft Excel	Office 2010 or Higher
Office Tools	Visual Studio 2010 Tools for Office Runtime

Figure 9: Client’s requirements

7.4. Access and Security

Apart from username/password, the system uses an action-based security model, to control access to the system’s capabilities and business objects. Roles are defined to correspond to the business functions within the enterprise, appropriate access is assigned to these roles, and then each user is assigned one or more roles according to the jobs he or she performs for the organization. Privileges can be set for users, roles, and books.

Hitachi Energy

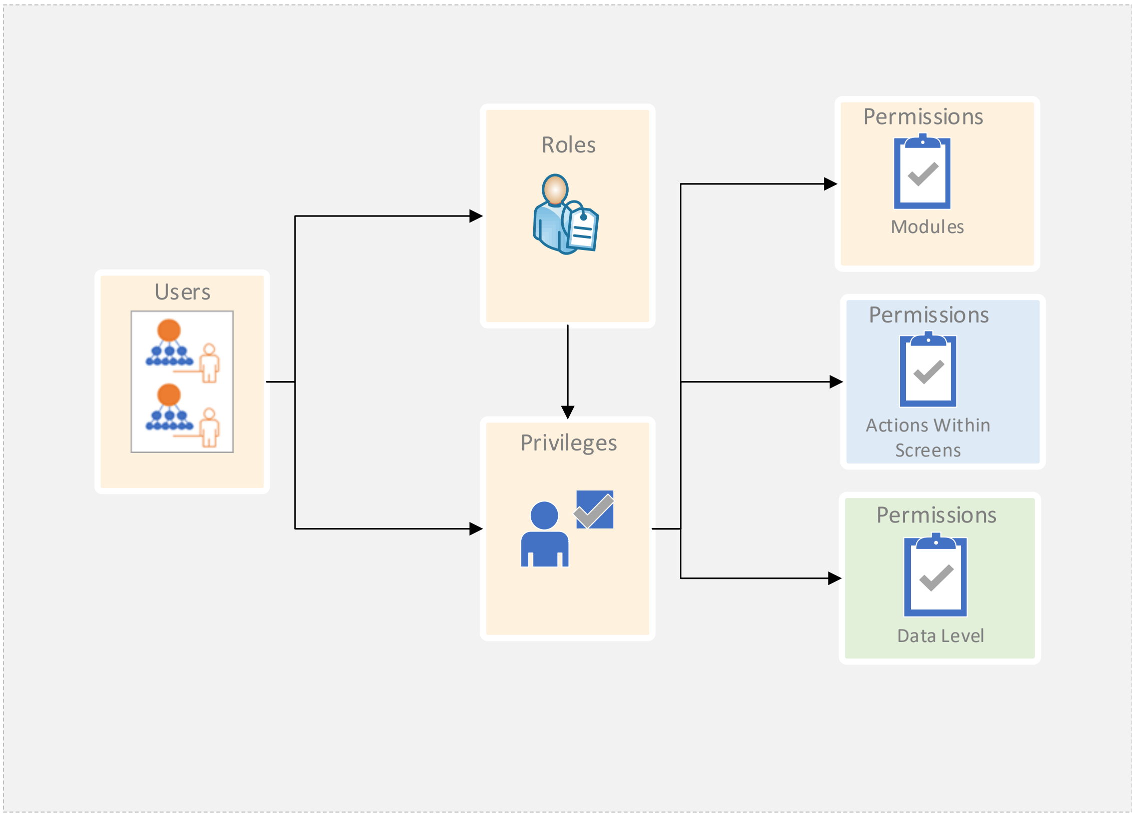


Figure 10: Access and security

Passwords are encrypted in the database. Data is not encrypted in the database. However, data transferred through the network is encrypted and secured based on HTTPS protocol.

7.5. Integration Capabilities

Hitachi Energy has invested a lot in integration methods and the system supports several integration approaches. Our system easily integrates with other applications and data sources using various methodologies using flat files, APIs, database access, the middle layer to Web Services.

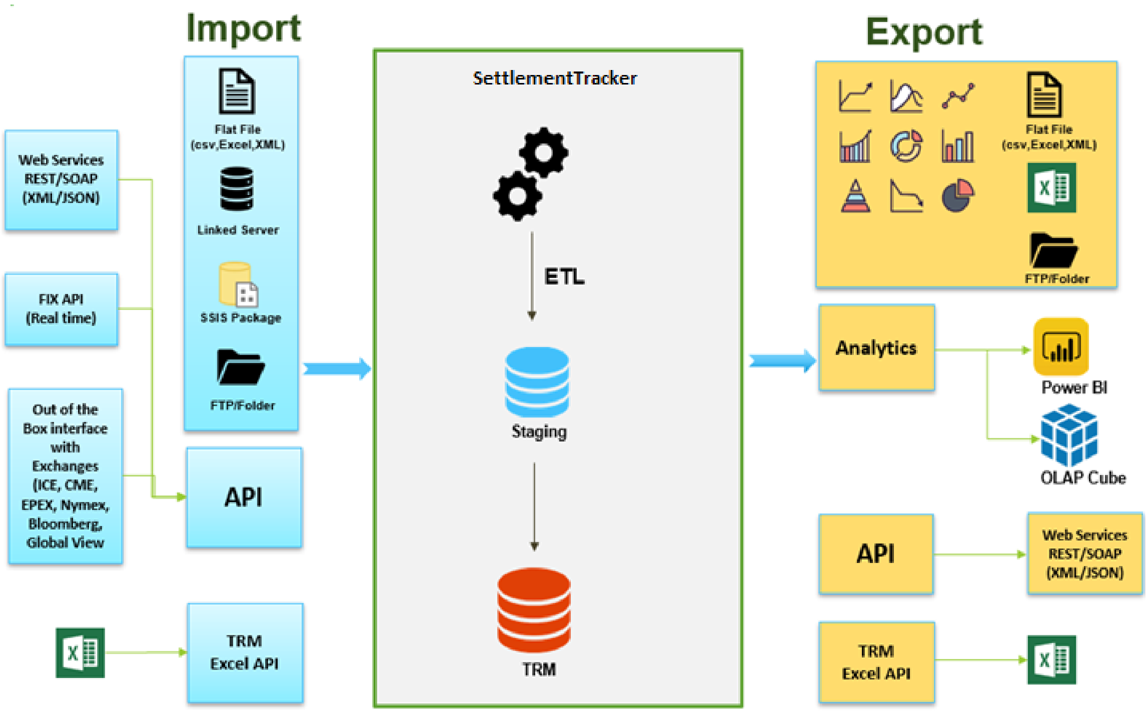


Figure 11: Integration capabilities

Please find below supported interface formats:

Inbound	Outbound
✓ Flat file ✓ Excel ✓ XML ✓ EDI ✓ REST-based web services ✓ SOAP-based web services ✓ FTP ✓ SSIS (SQL Server integration services) ✓ Linked Server ✓ ODBC ✓ FIX API	✓ Flat file ✓ Excel ✓ XML ✓ REST-based web services: TRM can export data via REST/web services. Any reports available in the application can easily be exported using the API and the attributes are customizable ✓ SOAP-based web services ✓ FTP ✓ Linked Server ✓ ODBC ✓ FIX API

Figure 12: Supported interface formats

8. Implementation (Services and Training)

This section provides details on the implementation aspects of the offered solution, user training and post-implementation support.

8.1. General Implementation Approach

We recommend an agile/iterative implementation process. Note that while we prefer this approach, we are happy to consider other processes that best fit ARRASS's requirements.

Our agile process will consist of multiple iterations. Typically, each of the iterations (or Sprint) will last approximately 2-4 weeks and should yield as much tangible and visible software delivery as possible. The beginning iterations should focus on proof of concepts of any "unknown" technological or functional aspects involved in the development and delivery of the software. This will help us minimise risks and increase certainty.

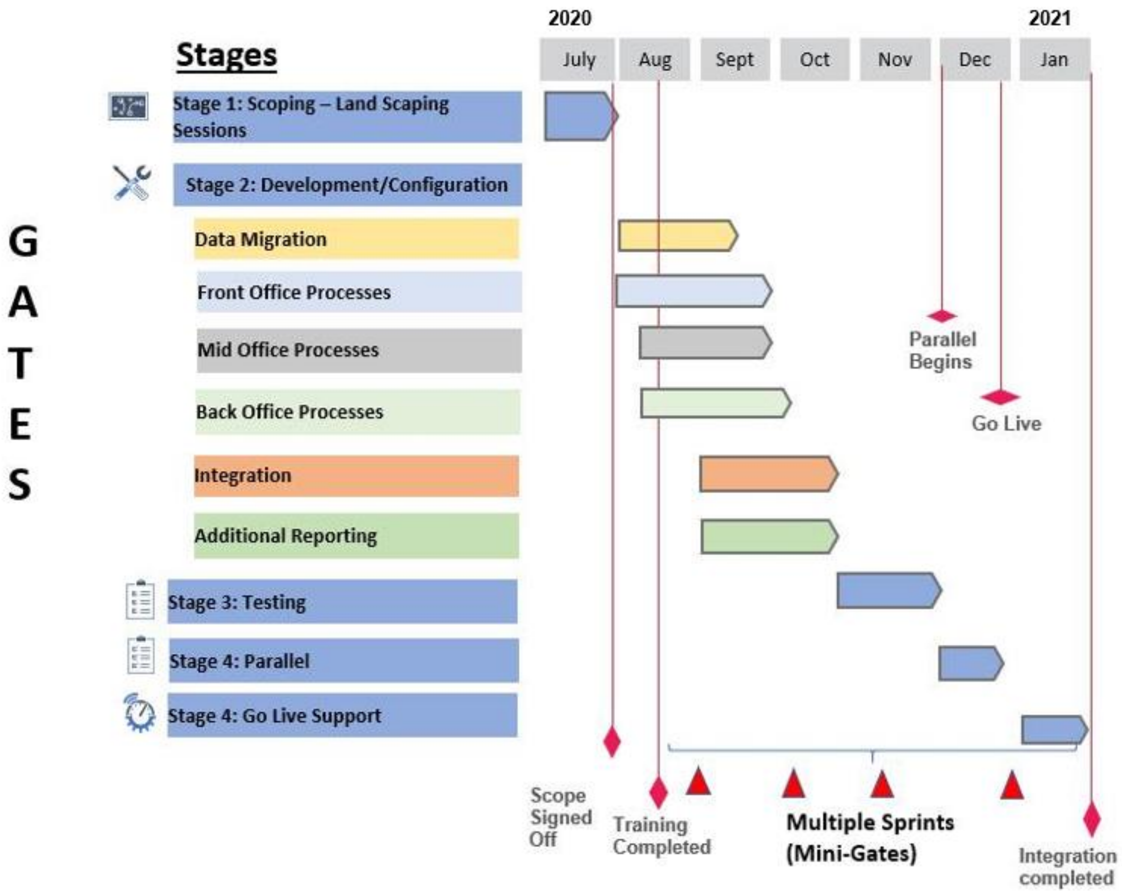


Figure 13: General implementation approach

We will start with a high-level project scoping phase, also known as the Project Landscaping phase. This initial phase will then be followed by multiple, additional iterations to implement the solution. Each of these multiple iterations has four steps: (1) Inception, (2) Elaboration, (3) Construction and (4) Transition. After completing each iteration, we then review its results to correct mistakes and identify potential areas for improvement. This ensures we are constantly trying to improve the process for the remaining iterations. Good project and configuration management are crucial to the success of the overall process. The diagram below demonstrates the 4 key attributes that contribute to the success of our implementation process lifecycle.

Inception	Elaboration	Construction		Transition
Plan	Architect	Configure	Verify & Train	Deploy/Turnover
• Project Team Assembled • Project Kick-off • Prepare project plan & scope	• Scoping / Functional overviews • Integration & reports discovery • Design workshops and proof of concepts • Architectural considerations • Review & Approve Design • Update project plan	• SettlementTracker installation • Baseline data • Core team training • Incremental deliveries • Integration API's • Unit Testing	• End to end Testing • User acceptance testing • End user training	• Production data loading • Go live check • Helpdesk procedures • Transition to next phase
• Project Management				

Figure 14: Attributes of the implementation process lifecycle

Plan Stage - The objectives of the Plan stage are to define the overall project scope and timelines for the initial go-live and to develop the procedures and mechanisms to plan and control the project. This stage sets the overall direction and approach for managing the project and is critical to its success. This phase formally documents the overall project scope and ensures its execution through a project plan. The Plan stage also defines the team members, roles and responsibilities, and the communication plan that will be used throughout the project.

Architect Stage – The Architect state of a SettlementTracker implementation allows the ARRASS and Hitachi Energy project team members to find a common understanding of ARRASS's business needs and to analyse how to best implement SettlementTracker to meet those needs. This stage is organised into two components: discovery and analysis. The discovery component gathers information on the current business practices and documents the current process and business objectives. The analysis component documents business requirements, designs proposed processes, identify gaps, and analyses how SettlementTracker will be configured and implemented to meet the business requirements.

Configure and Proof of Concept Stage – The objectives of the Configure & PoC stage are to complete the configuration of the SettlementTracker solution based on the business process design specification and the architectural specifications. All application configurations are completed, data conversation for both setup and user-level information are executed, integration configuration is completed, and the system is prepared for the test stage. During this stage, we also complete change management, training, and communication planning. The diagram below shows the repetitive nature of this stage within the overall project approach.

Verification and Train Stage – During this stage, the team ensures the configured SettlementTracker system sufficiently meets the needs of the business. The Verification & Train stage consists of three separate efforts 1) end-to-end system testing, 2) user acceptance testing, and 3) end-user training.

Deploy/Turnover stage – This phase includes the steps necessary to move the SettlementTracker solution into production with the features/functionalities in the scope of the project. The project team will coordinate the move to production with the customers' normal production change control process and effectively transition ARRASS to production support.

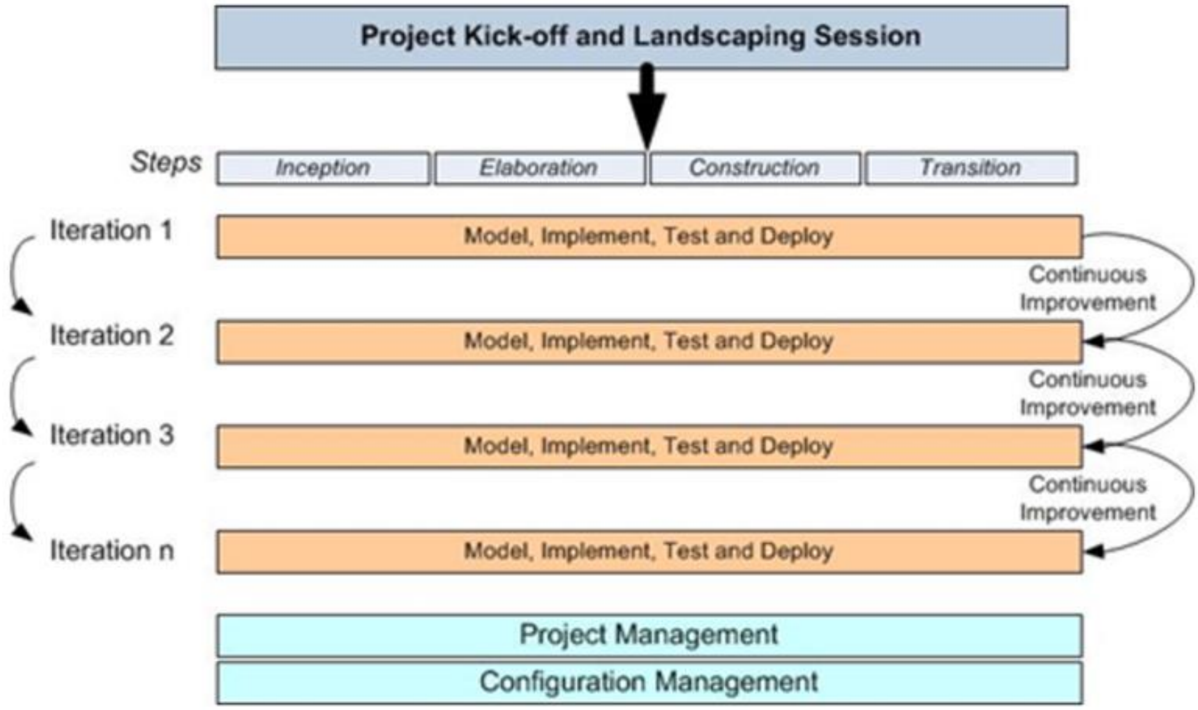


Figure 15: Implementation process lifecycle

8.2. Project Organisation

Services are provided under ARRASS leadership and project management. Hitachi Energy will provide a Project Coordinator, whose role is sole to oversee the implementation of the project deliverables at Hitachi Energy concerning time, budget, and quality. This will provide effective management which facilitates activities in a timely manner, ownership of the project, and a cost-effective approach.
We identify the following project team responsibilities:

Sponsor/Management: ARRASS management will be responsible for project approval, financing, and acceptance. Specifically, they will be responsible for approving project schedules, providing necessary resources, and accepting the project deliverable at milestones and completion.

Hitachi Energy Project Team: This team will be responsible for conducting project activities, providing progress reports, and producing the deliverables.

ARRASS Project Team: This team will be responsible for project management, and for making the ARRASS personnel, processes, tools, and local infrastructure available to the Hitachi Energy project team as needed during the project.

This organization will provide effective management to facilitate activities in a timely manner. It will ensure visible sponsorship and ownership of the project.

8.3. Resources

ARRASS Team

In the table below are the type of resources required from ARRASS. One person may assume multiple roles. It might be helpful to designate backup contacts in case the primary contact is not available. Note that this is a high-level estimate and will be revised depending on resource availability and tasks outlined in a detailed project plan.

ARRASS	Description
Project Manager	About 50% responsible for overall SPM & PASS project

Project Sponsor	5% Ultimate decision-maker
Back Office – Process Owner	50% Assist in finalising requirements and sign-off and overall ownership of back-office processes
IT Interface Support	25% - Provides support related to external data sources
IT Support	25% - IT liaison and interface support. This role will support internet connectivity, browser support, and other technical support.

Figure 16: ARRASS Team

Hitachi Energy team

Hitachi envisions a team consisting of a Project Coordinator, a Business Analyst and in the background 2-3 resources, that are not customer-facing / having direct customer contact.

Hitachi Energy team		
Name	**Role**	**Description**
TBD	Project Coordinator	Manage deliverables from a time and budget perspective
TBD	Senior Business Analyst(s)	Configuration, testing, interfaces, training
TBD	Subject Matter Expert	Initial scoping and solution architecting
Off-Site	Technical Architect	Technical
Off-Site	Business Analysts	Support as directed
Off-site	Developers	Support as directed
	Commercial	Account Executive, Commercial and Sales Support

Figure 17: Hitachi Energy team

The Hitachi Energy team may draw from local resources for project management as well as specialists from Hitachi staff that have directly implemented similar settlement solutions.

Hitachi Energy team members have received college education ranging from economics to computer science and business administration. They bring years of ETRM project experience with customers in Europe.

8.4. Recommended Software Tools for Project

These tools may not be provided by the customer and not by Hitachi Energy. We have worked with these tools during previous projects:

- Project Management
  - JIRA: For tracking of agile project management (sprints, impediments, actions)
  - Confluence: For managing/archiving documents, manual, use cases etc.
- Communication
  - Email
  - Screensharing tools, like Microsoft Teams, Skype for remote meetings & demos.
- Office Tools
  - Microsoft Excel: Other than for obvious reasons, this is also used for Data import functionality.

8.5. Risks and Mitigations

Enough time and budget must be allocated to the business requirements gathering activities to ensure that both Hitachi Energy and the ARRASS project teams have a common and clear understanding of the system application requirements and expectations. The risk assessment will be continuously monitored throughout the life of the project, with periodical assessments in the status report. Because mitigation approaches must be agreed upon by project leadership (based on the assessed impact of the risk, the project's ability to accept the risk, and the feasibility of mitigating the risk), it is necessary to allocate time into each status update meeting, dedicated to identifying new risk and discussing mitigation strategies, provided ARRASS has chosen to include this level of project management services from Hitachi Energy.

Risk	Mitigation Strategy
Estimated Project Schedule	The project approach includes the creation of a comprehensive project timeline with frequent baseline reviews
Scope / Deliverables management	Utilise a project management tool (Customer-provided) that allows management sprints, backlog etc.
Project Scope Creep	The project approach includes producing a detailed approved requirements document that will be strictly followed by Hitachi Energy. Anything not in this document is considered out-of-scope.
Lack of resources and buy-in to changes will delay the project.	No doubt, this project will require changes in the ways the ARRASS resources are accustomed to completing their tasks. Not everyone welcomes change. The lack of buy-in will delay this project and add costs. ARRASS will have someone to focus on Change Management who will champion the changes. Management should be committed throughout the project.
Undue and drawn-out acceptance testing	Follow a method whereby Hitachi Energy demonstrates the delivered functionality, after which the functionality will come under support and defects will be repaired under the maintenance agreement.
Decision-making authority	Based on a Statement of Work a list of team members is designated that must be able to make decisions related to their areas of the project to avoid delays.
End-User acceptance of an integrated and automated enterprise system; aversion to change, the end-users lack of big-picture understanding failing to see how what they do in the system affects upstream and downstream operations. Recognition that a front-to-back system is not a spreadsheet.	Train core users to create system champions (train the trainer). Provide sufficient training before go-live, including insight into the workflow. Provide post-training services to optimise the benefits of the just acquired skills and knowledge, thereby maximising the success and return on investment of the Hitachi Energy system
Change management	Follow a process for changes when after the approval of the requirements document, designated functional project team members may initiate change requests. Each change request must be

Risk	Mitigation Strategy
	submitted in writing and detail an aspect of the system which either needs to be implemented differently than documented in the requirements or needs to be added to the requirements. The Project Manager will request that Hitachi Energy develop a cost estimate and proposed delivery schedule for the added or modified functionality. If funding is required, then upon authorization in writing and approval of funding by the Project Management Lead, and according to the delivery schedule, Hitachi Energy will commence work on the requested change.

Figure 18: Risks and mitigations

8.6. High-Level Project Timeline

We will provide a more definitive detailed project timeline during Scoping Phase.

The following is an indicative timeline based on our experience and the preliminary information of ARRASS's requirements. For that reason, this should be considered tentative and subject to ARRASS feedback and availability.

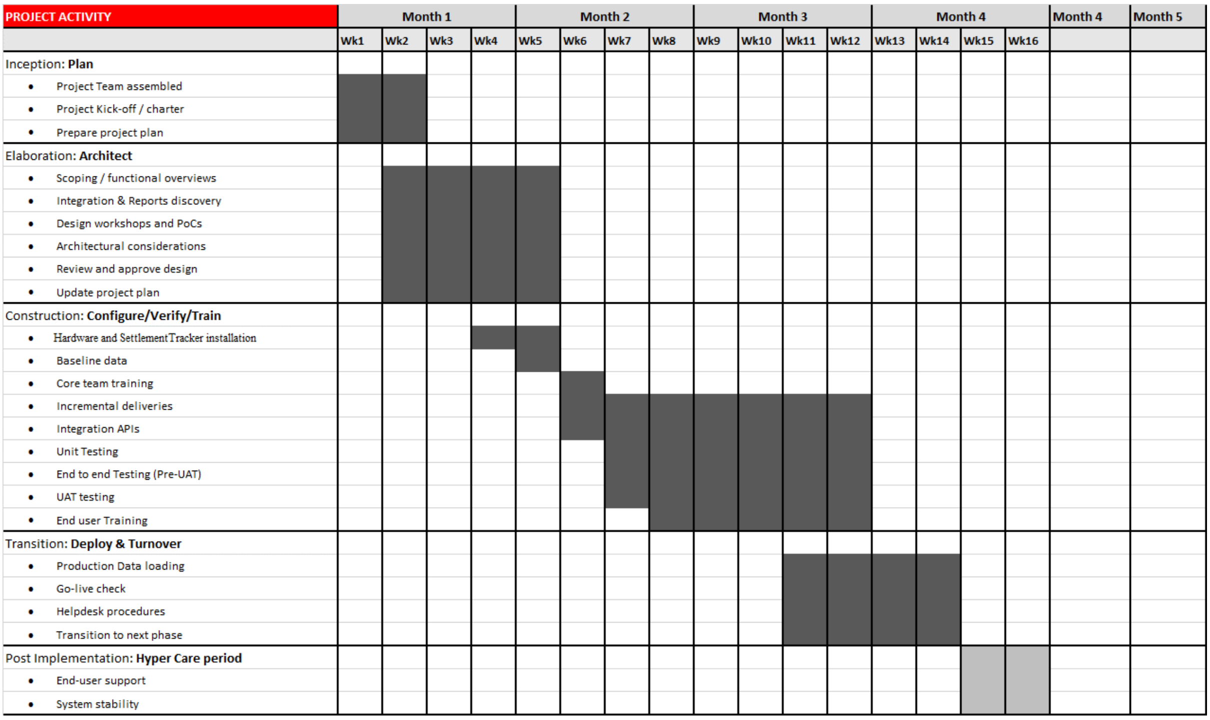


Figure 19: High-level project timeline

8.7. Training

Another benefit of our agile, iterative delivery process is the fact that the core team users get hands-on experience with the SettlementTracker software from the beginning of the project to the final go-live version. During implementation, they work side-by-side with Hitachi Energy staff (i.e., hands-on training). Most are ready to operate the software by the end of implementation which minimises additional training and allows ARRASS to quickly realise results.

Our training programs are hands-on and tailored around the skill set of the users, thus building confidence, and facilitating full utilization of the system's potential. Depending on the type of deployment, we can provide the following training sessions:

- Workshop with functional training to business users
- Administrator training for administrators/core users who will configure the system
- Technical training for those who are responsible for technical aspects of the system.

Training can be done onsite or remotely using web-meeting methods. Hitachi Energy does offer refresher courses as needed by our clients by request. Hitachi Energy does offer skill upgrades and new employee training by request. We maintain both functional and technical user manuals.

8.8. Post Implementation – Technical Support

After the production-ready software has been accepted and allows ARRASS to go live, a Hyper Care phase may be entered. Our implementation team resources are still involved to provide end-user support and ensure system stability.

Our standard technical support is 8 AM-5 PM during business days. Technical support and maintenance updates are provided under the license agreement.

Technical support is provided based on Hitachi Energy's Services Level Agreement (SLA) which provides detailed information on our resolution process, response times, issue escalation protocol and document resolutions.

While 24/7 dedicated support is optional, we rarely sell this service.

We employ a distributed software support network in the U.S., Europe/Middle East, and Asia. This network allows us to extend our support beyond standard business hours when required. This allows for exceptional customer service and has been a big reason for our success—and our clients, along with other industry professionals, agree. For six years running, we have been ranked in the Top Five for customer support/services.

In many cases, customers are finding their questions answered and issues resolved within 24 hours. We offer call-in numbers that are always available during work hours. Our standard technical support service includes up to eight hours of Phone Technical Support per Month (Calls are measured by 15min increments) Mon-Fri (8 AM to 5 PM Central European Time).

Given the effective and efficient Email/Ticket System communication, the 8 hours of phone support is therefore available for urgent matters, while the rare emergencies - the events where there is a loss of service, either the service is not accessible or a most business-critical function is non-functional - wouldn't be counted those calls against the eight hours.

Support vs Warranty

Hitachi Energy license and support agreement distinguishes between maintenance and product warranty. All products are licensed with one year of standard technical support and maintenance. A 90-day warranty period is to establish the software performs as described. Errors and defects in the software are covered by and resolved under the maintenance agreement.

Updates and Enhancements

Technical support and maintenance on software customization that become part of the core-commercial system may be covered by a proportional increase of the support fee to cover the enlarged scope of testing, verification, packaging, and deployment activities, or provide support on a time and material basis. If not, then support for custom software not part of the core-system is can be provided on a time and material basis.

We are constantly improving our solution usually to complement or satisfy our clients' requests or to satisfy functional gaps. Our release schedule is as follows a) Minor (maintenance) releases every 6 months b) Major releases every 2 years c) Hotfixes of critical issues on demand. Our maintenance and support agreement offers remote system review. Onsite visits can be negotiated for an additional cost.

The system provides a high degree of user-configuration ability and therefore the customer does not need to rely on Hitachi to make changes to deal templates, book structures, workflows, reports, invoice templates, flat-file data import rules and more.

9. Assumptions

This proposal is based on the following assumptions:

- Hitachi Energy will manage the day-to-day activities of its resources only and not those of either the Customer or the End Client or other vendors that the Customer or End Client might engage on the Project.
- Hitachi Energy's ability to deliver as per the milestones and performance measures may be constrained by dependent activities that are the responsibility of the Customer or the End Client or other third-party vendors, and hence Hitachi Energy will not be responsible for any such delays.
- Where Hitachi Energy is requested to engage in activities outside of the scope defined above (e.g., a variation to the environment build rollout strategy), these will be considered as a variation and related pricing will be supplied on request.
- The end Client will have the appropriate computer hardware and technical environment in place and will provide all required access prior to Hitachi Energy commencing work.
- If work is to be performed at End Client's facility, End Client will ensure adequate office facilities in close proximity to the designated members of the End Client staff assigned to work with Hitachi Energy on this project. Facilities for each consultant will include Internet access for accessing Hitachi Energy intranet using Hitachi Energy laptop computers.
- The End Client will ensure a VPN connection is available for Hitachi Energy resources for off-site connectivity for activities requiring remote access.
- The End Client is responsible for all software installation activities not explicitly included in this Proposal.
- Services will be provided as a combination of off-site and on-site.
- Hitachi Energy's delivery of the Services is dependent on: (i) Customer or End Client's timely and effective completion of Customer Responsibilities, (ii) the accuracy and completeness of the Customer Responsibilities, and (iii) timely decisions and approvals by Customer or End Client's management.
- End Client will provide sufficient, qualified, knowledgeable personnel capable of (i) performing Customer Responsibilities; (ii) making timely decisions; and (iii) participating in the project and cooperating with Hitachi Energy.
- If Customer or End Client does not perform Customer Responsibilities in a timely manner, Hitachi Energy may: (a) suspend the Services; and/or (c) terminate the contract.
- Any software and documentation provided under this proposal will be delivered electronically.
- Hitachi Energy Project Manager will provide a single point of contact between Customer and Hitachi Energy with regard to scope, schedule, and resources assigned to provide the Hitachi Energy services.
- Hitachi Energy resources will work under the direction of a Hitachi Energy Project Manager or designee.
- No Acceptance Test or Warranty is applicable unless specifically noted in this proposal.
- All software, professional services and technical support will be delivered in the English language only.
- This proposal is for the Hitachi Energy expertise and implementation services required to deliver a solution based on the SettlementTracker software. This proposal includes only an estimation and can be amended by Hitachi Energy. A final Statement of Work (SOW) document will be created

describing the full scope and more details during the contracting phase which will need to be reviewed and agreed by the Customer as a signed SOW.

10. Change Control of Services

In the event, Customer desires a change in the terms or conditions of proposed services, and if, in Hitachi Energy sole judgment, such change results in (i) additional services, (ii) a need for additional time to perform the services, or (iii) additional costs to perform the services, the following change control procedures will be applied:

- The customer shall prepare and submit to Hitachi Energy a written request for any change ("Change Request"). Hitachi Energy shall develop a written response ("Response") to any Change Request. The customer shall pay for all reasonable design work performed by Hitachi Energy to develop its Response on a time and materials basis at rates as defined herein.
- Hitachi Energy receives a Change Request, Hitachi Energy shall provide Customer with a Response either (i) offering to perform services consistent with the Change Request, (ii) proposing modifications to the Change Request, or (iii) rejecting the Change Request. Hitachi Energy Response shall include detailed information as to (i) the availability of consultants, and (ii) the impact, if any, on the completion of services, the delivery of any Deliverables, and the cost of the services.
- If Customer desires to implement a Change Request, Customer shall provide written authorization for Hitachi Energy to proceed with such Change Request upon the terms originally set forth therein or as modified by Hitachi Energy Response pursuant to subsection (b) above. Hitachi Energy shall prepare an amendment ("Change Order") including an estimate of the cost of the additional services contained in the Change, and if the Customer is in agreement, both parties shall execute the Change Order prior to commencement of any work.
- Upon receipt of the Change Order executed by both parties, Hitachi Energy shall promptly commence performance in accordance with the Change Order.
- Each Change Order shall constitute a formal modification to, shall be deemed incorporated into, and shall become a part of the applicable agreement.

11. Licensing Model

The Hitachi Energy software ‘SettlementTracker for SPM and PASS version’ is offered under a perpetual license with the provision of annual maintenance. This software requires a Microsoft SQL Server database License (perpetual) along with its maintenance.

Named Users

SettlementTracker for SPM/PASS is licensed based on Named Users with a maximum of 10 users. Additional Users can be purchased.

Environments included

The SettlementTracker software license includes 1 x Production, 1x User Acceptance/Test, and 1 Development environment.

12. Terms and Conditions

Terms and Conditions are provided along with the Commercial Offer.

13. Legal Note

This document contains a proposal that is subject to written contract. It is not a legal offer that is capable of acceptance by the recipient or by any other person, nor is it an invitation to treat / invitation to bargain that is intended to invite a legal offer that is capable of acceptance. The proposal is made on the basis that any transaction that arises from it will be governed by Hitachi Energy standard terms and will be subject to both parties executing a formal written contract. Hitachi Energy provides this document 'as is', without warranty of any kind, either expressed or implied, including, but not limited to, the implied warranty of merchantability or fitness for a particular purpose. Hitachi Energy may make changes or improvements in the equipment, software, or specifications described in this document at any time and without notice. Hitachi Energy has made every reasonable effort to ensure the accuracy of this document; however, it may contain technical inaccuracies or typographical errors. Hitachi Energy disclaims all responsibility for any labour, materials, or costs incurred by any person or party as a result of their use or reliance upon the content of this document. Hitachi Energy and its affiliated companies shall in no event be liable for any damages (including, but not limited to, consequential, indirect, or incidental, special damages or loss of profits, use or data) arising out of or in connection with this document or its use, even if such damages were foreseeable or Hitachi Energy has been informed of their potential occurrence.

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13.1. Confidentiality Statement

Arrass , Larsen & Toubro shall not (a) directly or indirectly disclose or cause to be disclosed, or otherwise transfer any Hitachi Energy Confidential Information to any third party; or (b) utilise Hitachi Energy Confidential Information for any purpose, except as expressly contemplated by the applicable RFI, RFP, RFQ or similar document. Arrass will limit the disclosure of Hitachi Energy Confidential Information, to Affiliates and Employees with a need to know and who have been advised of the confidential nature thereof, or third-party consultants with a need to know and who have been contractually obligated to maintain such confidentiality through signature of a written nondisclosure agreement acknowledging the non-disclosure obligations set forth herein; provided, however, Arrass will obtain Hitachi Energy's prior written consent before disclosing any Hitachi Energy confidential Information to any third party.

13.2. Change in Law

Notwithstanding anything in this Agreement to the contrary, if after submission of this Hitachi Energy bid or during the term of the Agreement there are any amendments, repeals or new laws, ordinances, statutes, rules, regulations, orders, by-laws, codes or decrees (including changes in work permits, visa requirements, taxation and import/export), including but not limited to the Customer or Hitachi Energy country, which affect the performance of the Agreement including but not limited to cost and/or time, an equitable adjustment to the Agreement shall be made – as notified and based on documentation provided by Hitachi Energy – to: (i) the date of delivery or completion; (ii) the price, which shall be increased to compensate Seller for any increase in costs; or (iii) any other Hitachi Energy commitments or obligations taking into account the impact of such change in law.

Hitachi Energy